

SAPIENZA
UNIVERSITÀ DI ROMA

Programmazione lineare

Catalogo dei problemi

I problemi di programmazione lineare usati negli esempi e negli esercizi di tutto il corso, citati con la loro sigla P0–P86: problemi con due e tre variabili, divisi in quattro tipi (A: termini noti non negativi; B: simplesso duale; C: serve la Fase 0; D: problemi illimitati, inammissibili o con ottimi multipli). Per ciascuno il problema e la figura della regione ammissibile.

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Versione online e materiale scaricabile:
fabiofurini.github.io/programmazione-lineare/
github.com/fabiofurini/programmazione-lineare

Anno accademico 2026–2027

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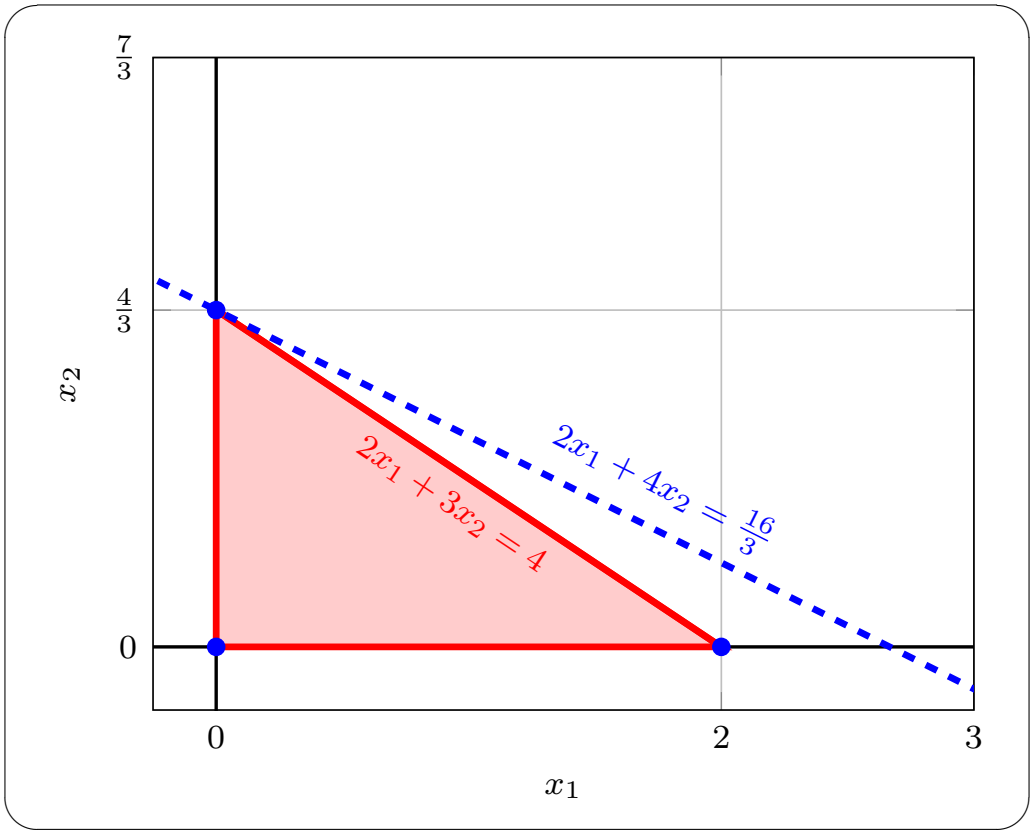
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Capitolo 1

Problemi LP con due variabili – Tipo A

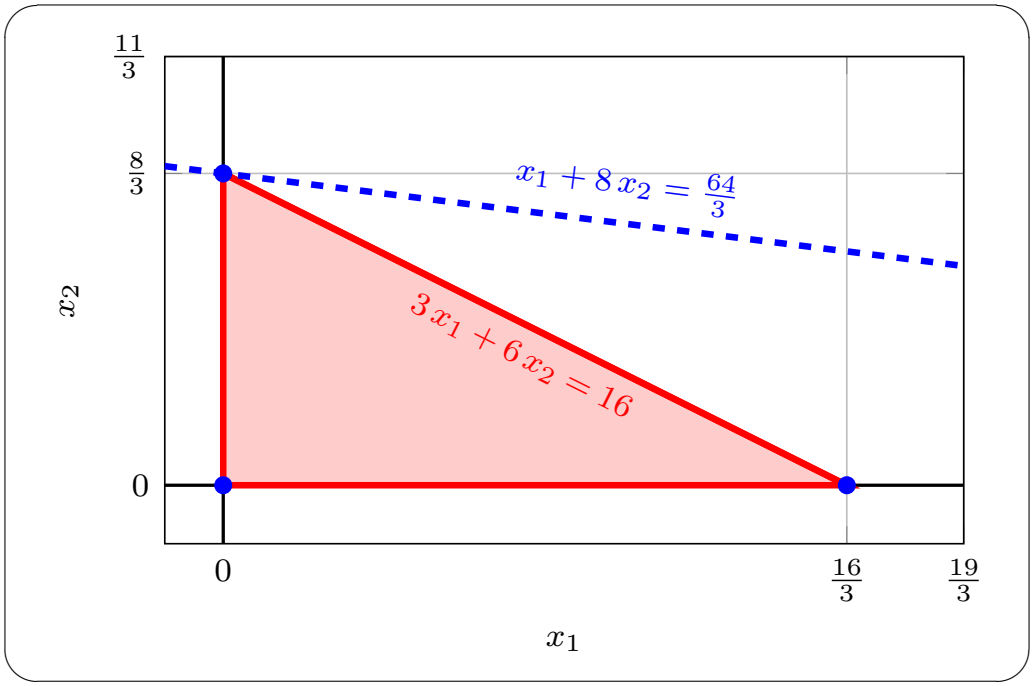
1.1 Problema LP $P0$

$$
\begin{array}{rrcrcrcl}
\max & 2 & x_1 & + & 4 & x_2 & & \\
\text{soggetto a} & 2 & x_1 & + & 3 & x_2 & \leq & 4 \\
 & & x_1, & & & x_2 & \geq & 0
\end{array}
$$



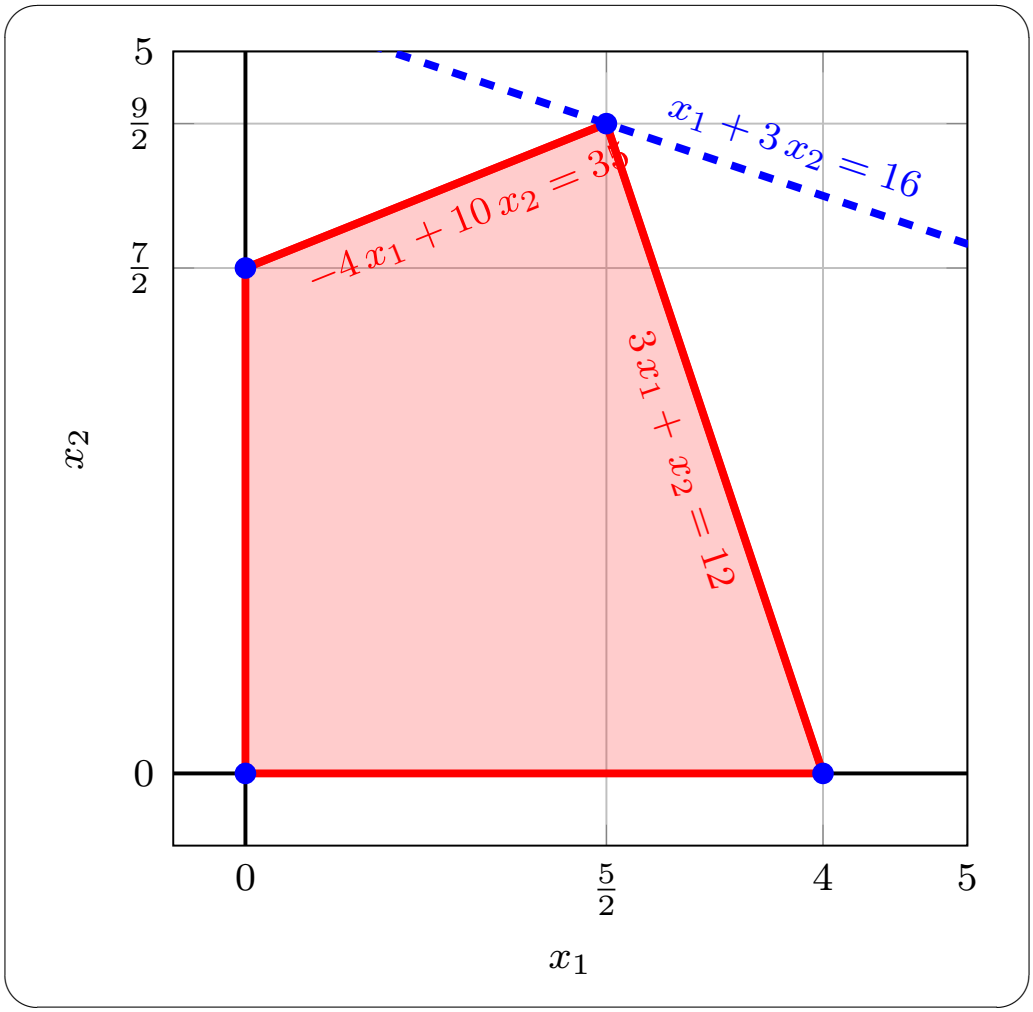
1.2 Problema LP $P1$

$$
\begin{array}{lrcrcrl}
\max & x_1 & + & 8 & x_2 & & \\
\text{soggetto a} & 3\ x_1 & + & 6 & x_2 & \leq & 16 \\
& x_1, & & & x_2 & \geq & 0
\end{array}
$$



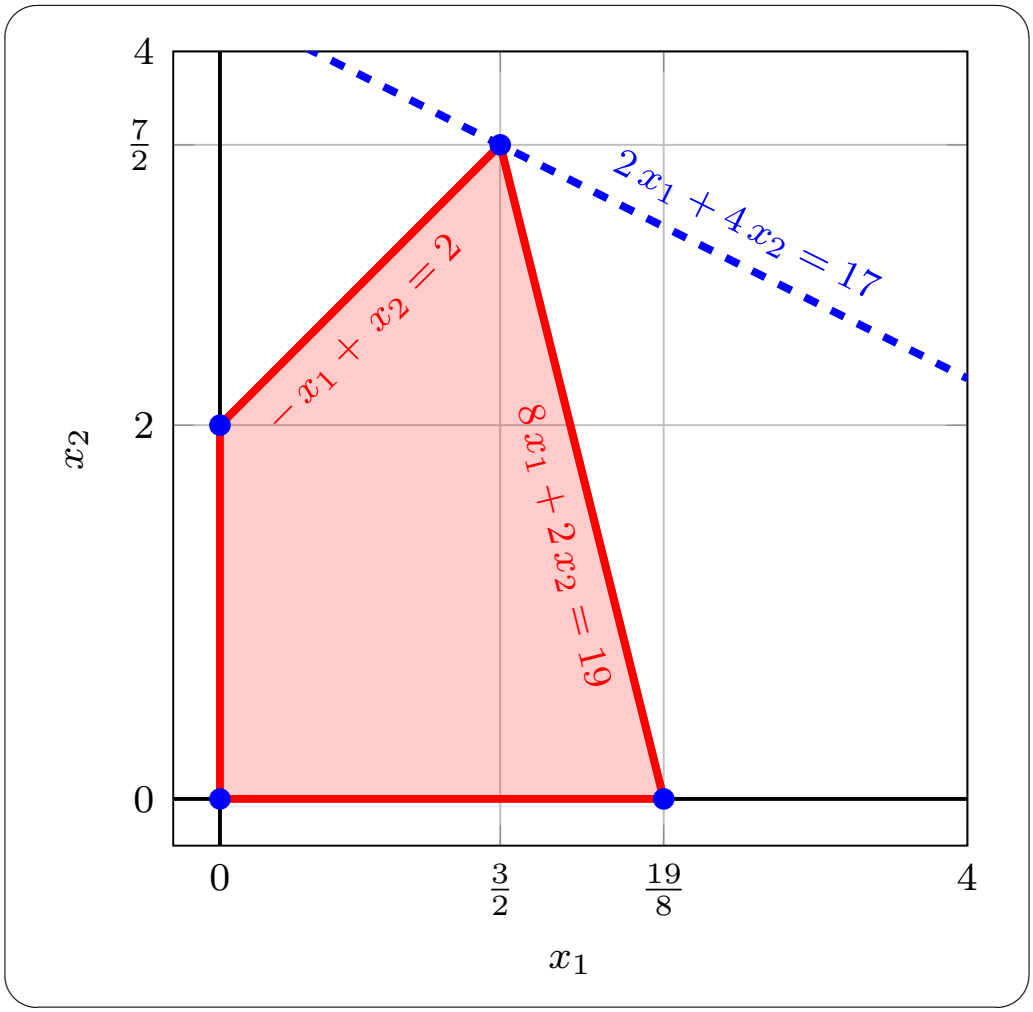
1.3 Problema LP *P2*

$$
\begin{array}{lrcrcrl}
\max & x_1 & + & 3 & x_2 & & \\
\text{soggetto a} & -4\,x_1 & + & 10 & x_2 & \leq & 35 \\
& 3\,x_1 & + & & x_2 & \leq & 12 \\
& x_1, & & & x_2 & \geq & 0
\end{array}
$$



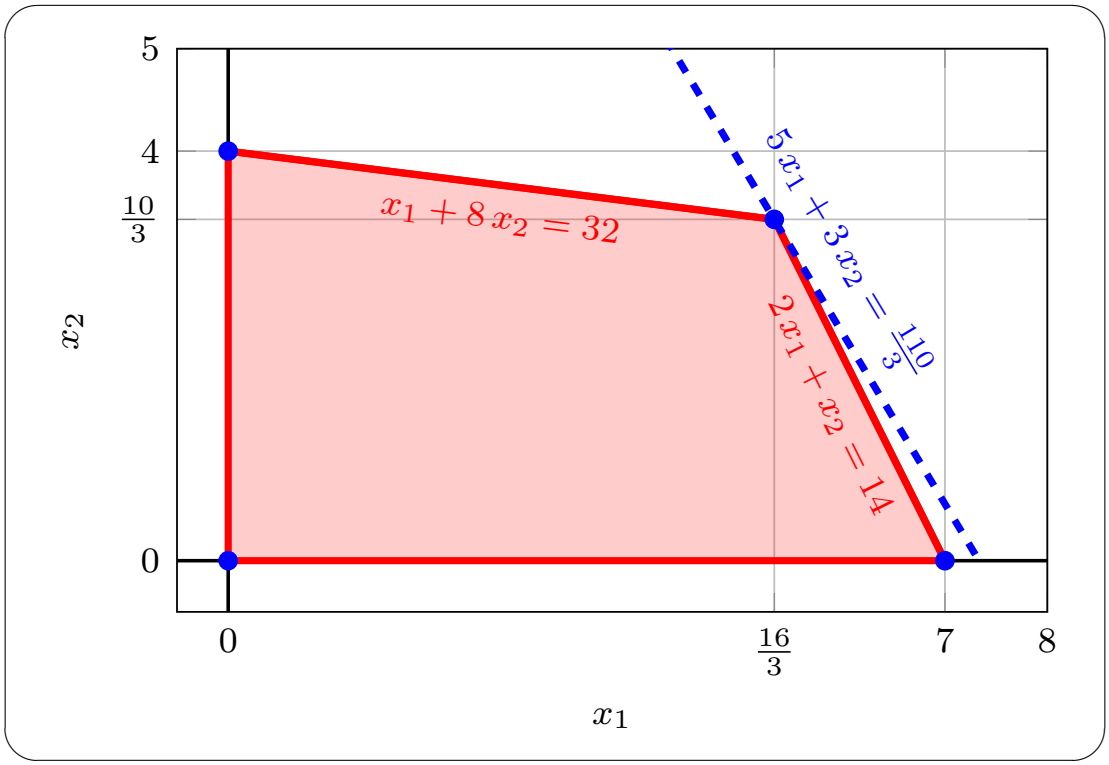
1.4 Problema LP *P3*

$$
\begin{array}{rrcrcrcl}
\max & 2 & x_1 & + & 4 & x_2 & & \\
\text{soggetto a} & - & x_1 & + & & x_2 & \leq & 2 \\
& 8 & x_1 & + & 2 & x_2 & \leq & 19 \\
& & x_1, & & & x_2 & \geq & 0
\end{array}
$$



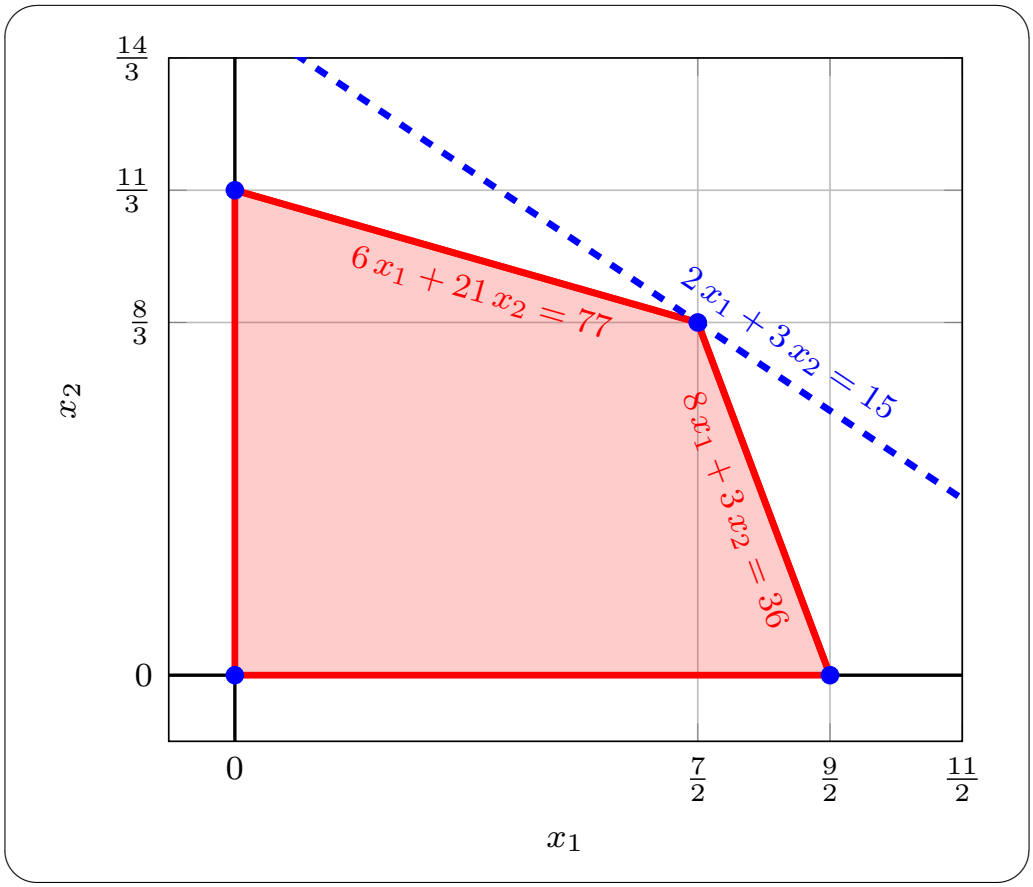
1.5 Problema LP *P4*

$$
\begin{array}{rrcrcrl}
\max & 5\,x_1 & + & 3 & x_2 & & \\
\text{soggetto a} & 2\,x_1 & + & & x_2 & \leq & 14 \\
& x_1 & + & 8 & x_2 & \leq & 32 \\
& x_1, & & & x_2 & \geq & 0
\end{array}
$$



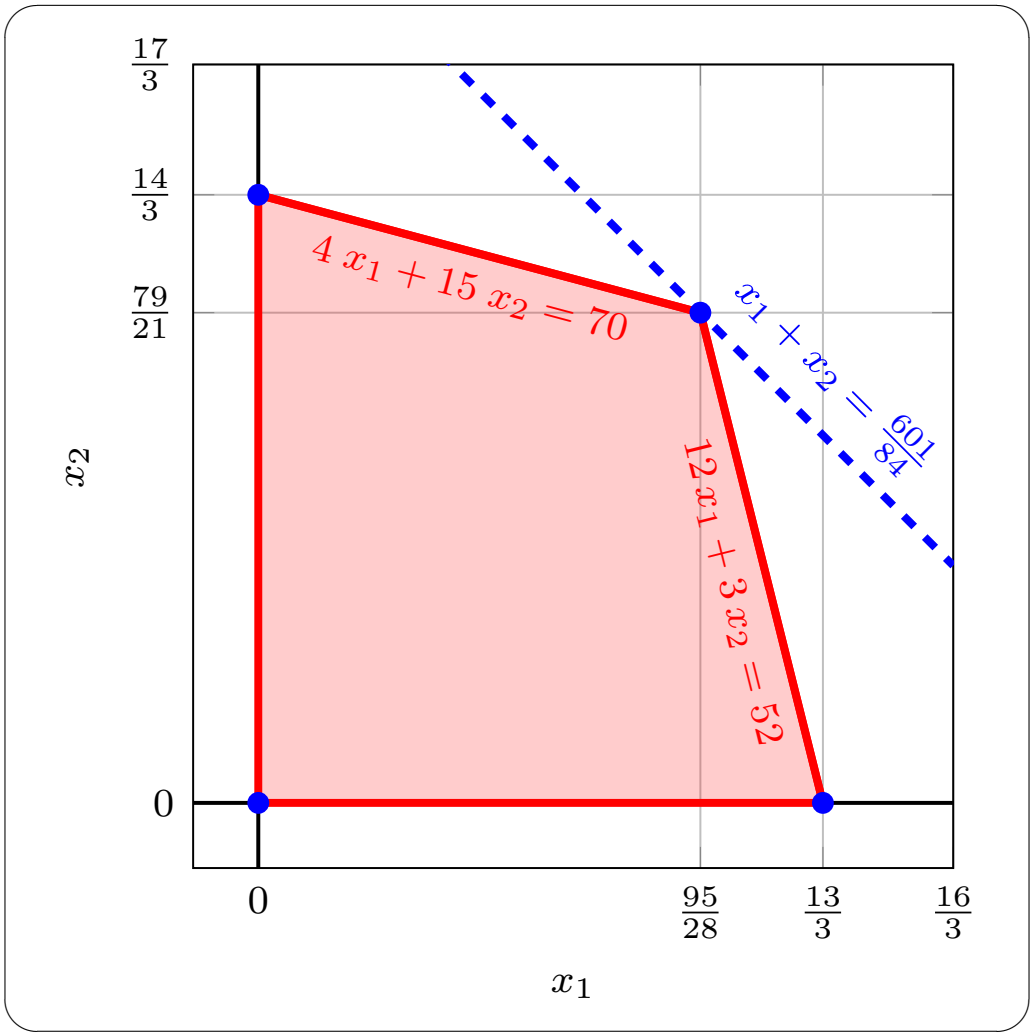
1.6 Problema LP $P5$

$$
\begin{array}{rrcrcrr}
\max & 2\,x_1 & + & 3\,x_2 & & \\
\text{soggetto a} & 6\,x_1 & + & 21\,x_2 & \leq & 77 \\
& 8\,x_1 & + & 3\,x_2 & \leq & 36 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



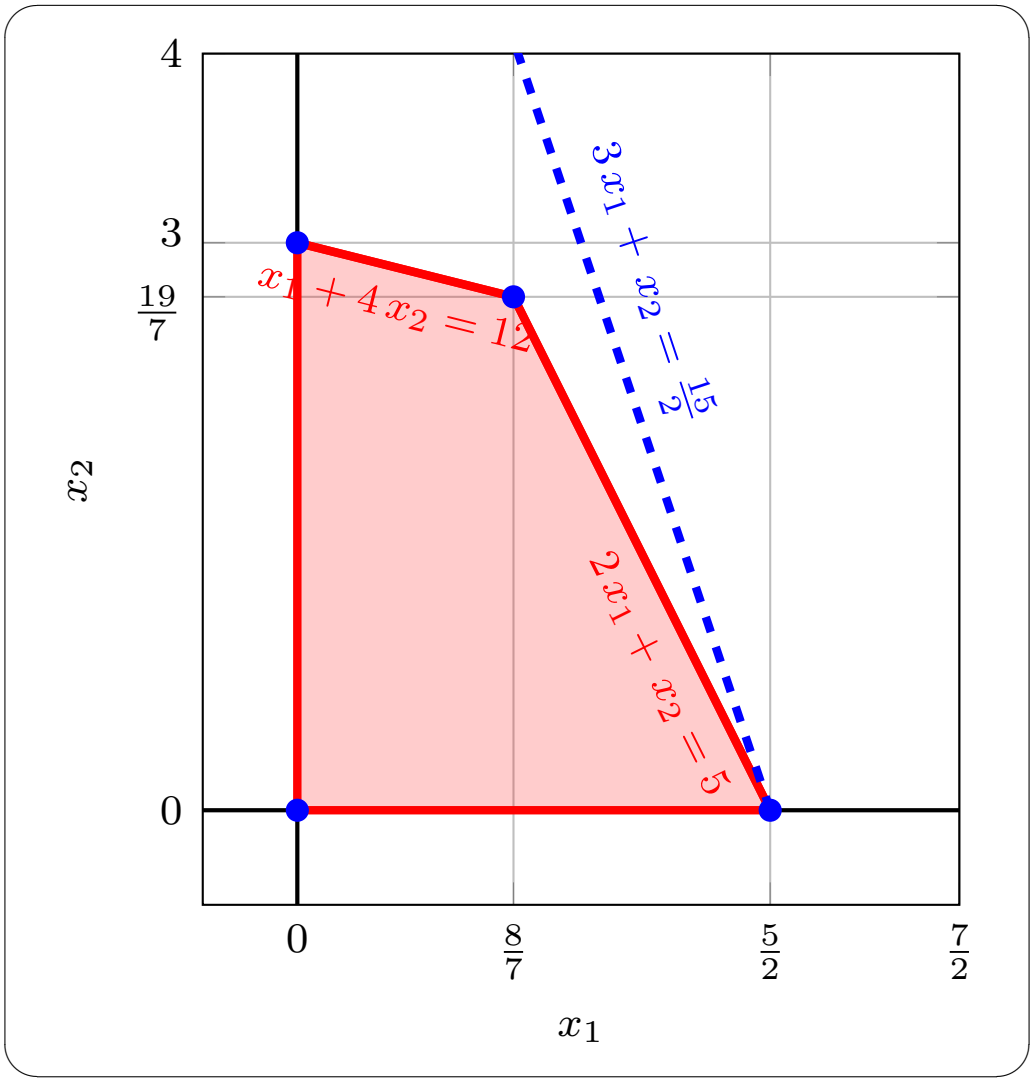
1.7 Problema ILP *P6*

$$
\begin{array}{rrcrcrl}
\max & x_1 & + & & x_2 & & \\
\text{soggetto a} & 4\,x_1 & + & 15 & x_2 & \leq & 70 \\
& 12\,x_1 & + & 3 & x_2 & \leq & 52 \\
& x_1, & & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



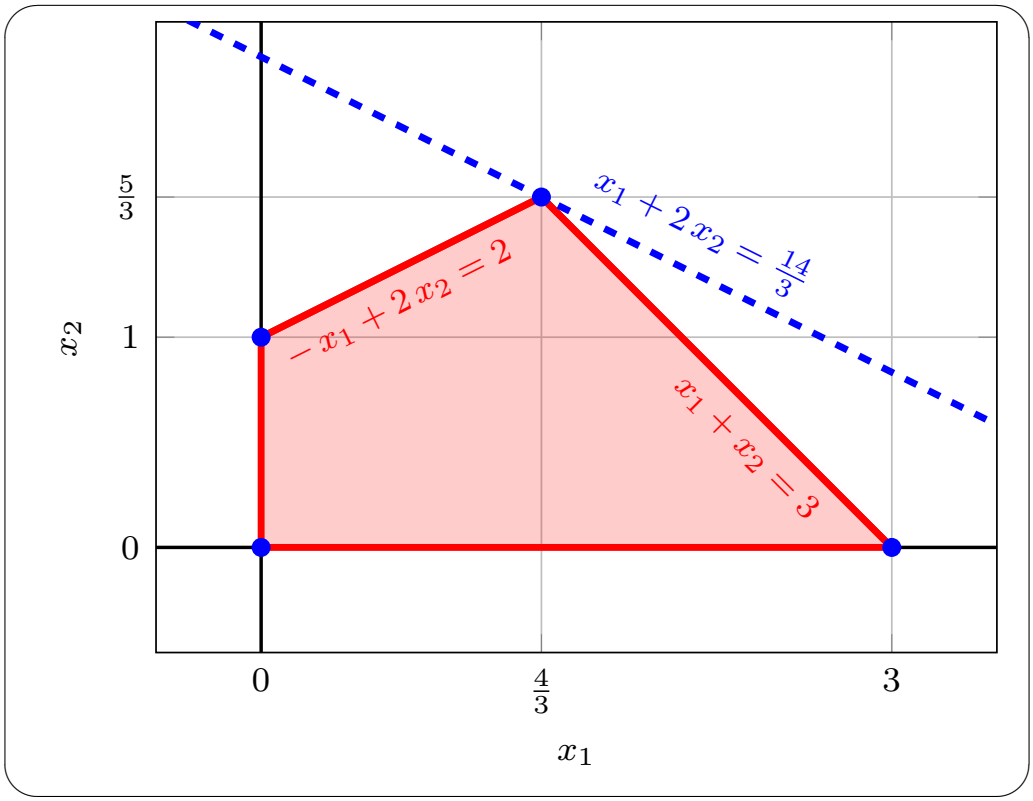
1.8 Problema LP *P7*

$$
\begin{array}{rrrrrrrr}
\max & 3 & x_1 & + & & x_2 & & \\
\text{soggetto a} & & x_1 & + & 4 & x_2 & \leq & 12 \\
& 2 & x_1 & + & & x_2 & \leq & 5 \\
& & x_1, & & & x_2 & \geq & 0
\end{array}
$$



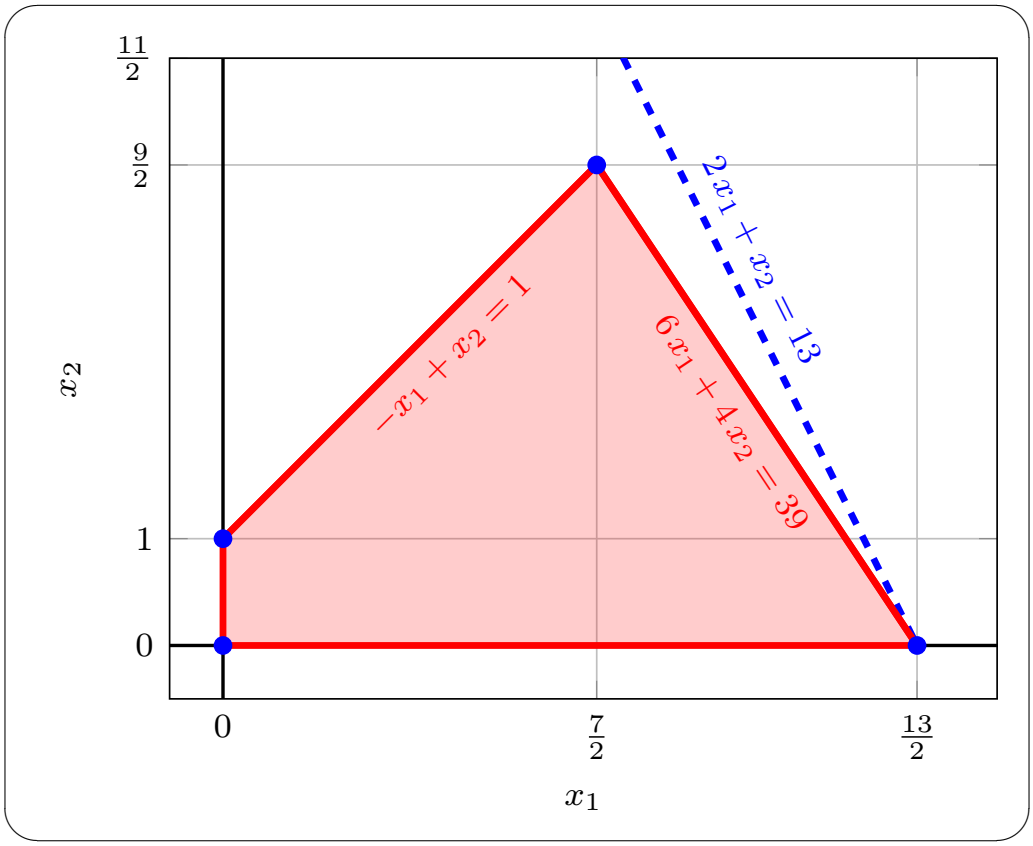
1.9 Problema LP *P8*

$$
\begin{array}{lrcrcl}
\max & x_1 & + & 2\ x_2 & & \\
\text{soggetto a} & -\ x_1 & + & 2\ x_2 & \leq & 2 \\
& x_1 & + & x_2 & \leq & 3 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



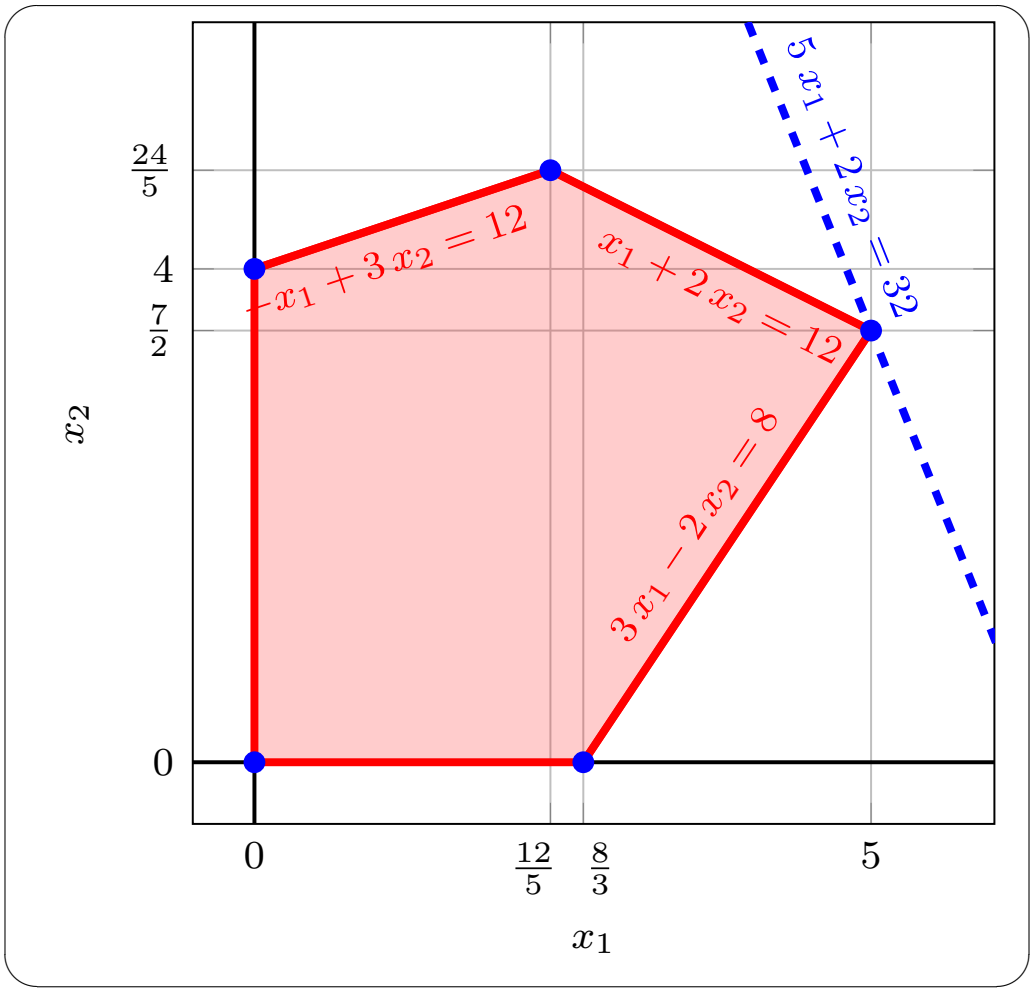
1.10 Problema LP *P9*

$$
\begin{array}{rrcrcrcr}
\max & 2 & x_1 & + & & x_2 & & \\
\text{soggetto a} & - & x_1 & + & & x_2 & \leq & 1 \\
& 6 & x_1 & + & 4 & x_2 & \leq & 39 \\
& & x_1, & & & x_2 & \geq & 0
\end{array}
$$



1.11 Problema LP *P10*

$$
\begin{array}{rrrrrrrr}
\max & 5 & x_1 & + & 2 & x_2 & & \\
\text{soggetto a} & - & x_1 & + & 3 & x_2 & \leq & 12 \\
 & & x_1 & + & 2 & x_2 & \leq & 12 \\
 & 3 & x_1 & - & 2 & x_2 & \leq & 8 \\
 & & x_1, & & & x_2 & \geq & 0
\end{array}
$$

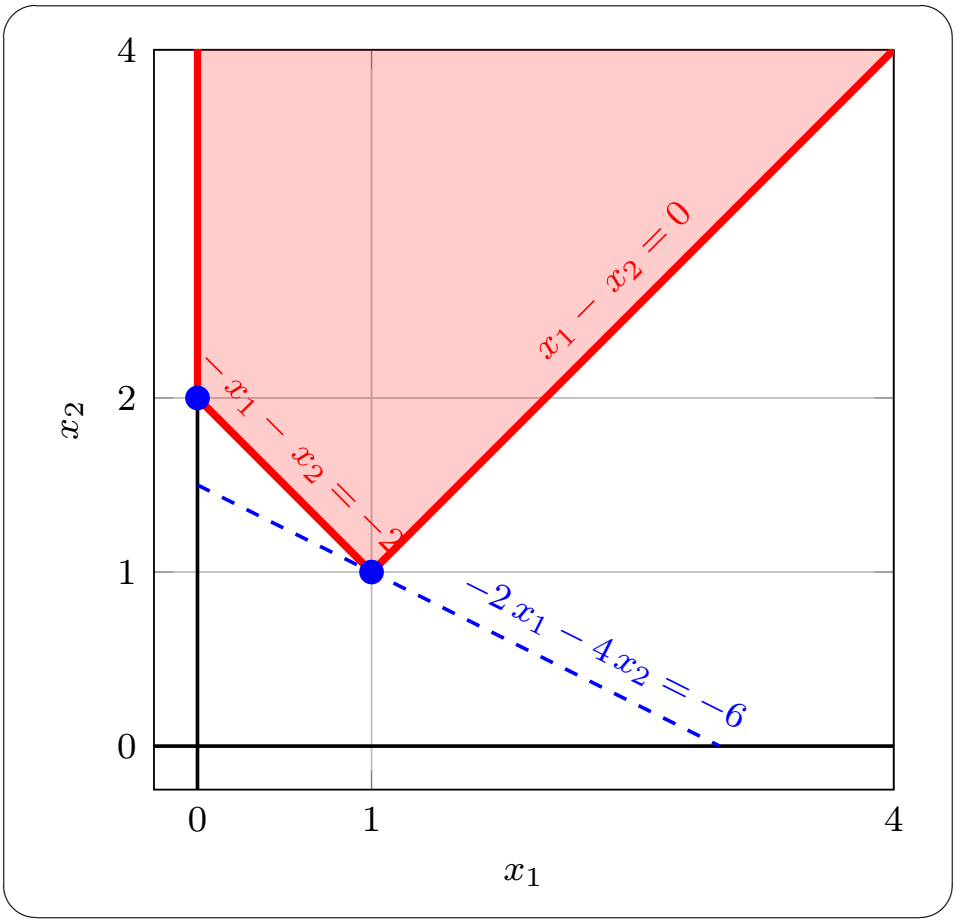


Capitolo 2

Problemi LP con due variabili – Tipo B

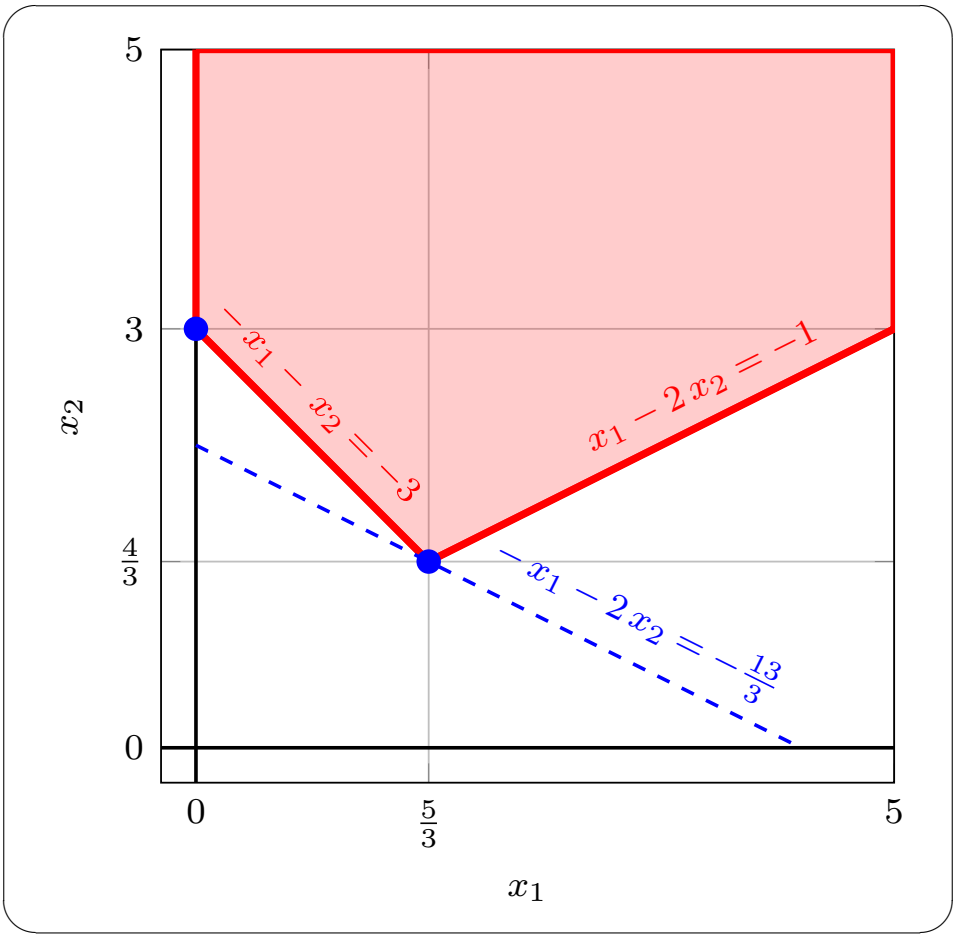
2.1 Problema LP *P11*

$$
\begin{array}{rrcrcr}
\max & -2\,x_1 & - & 4\,x_2 & & \\
\text{soggetto a} & -\,x_1 & - & x_2 & \leq & -2 \\
 & x_1 & - & x_2 & \leq & 0 \\
 & x_1, & & x_2 & \geq & 0
\end{array}
$$



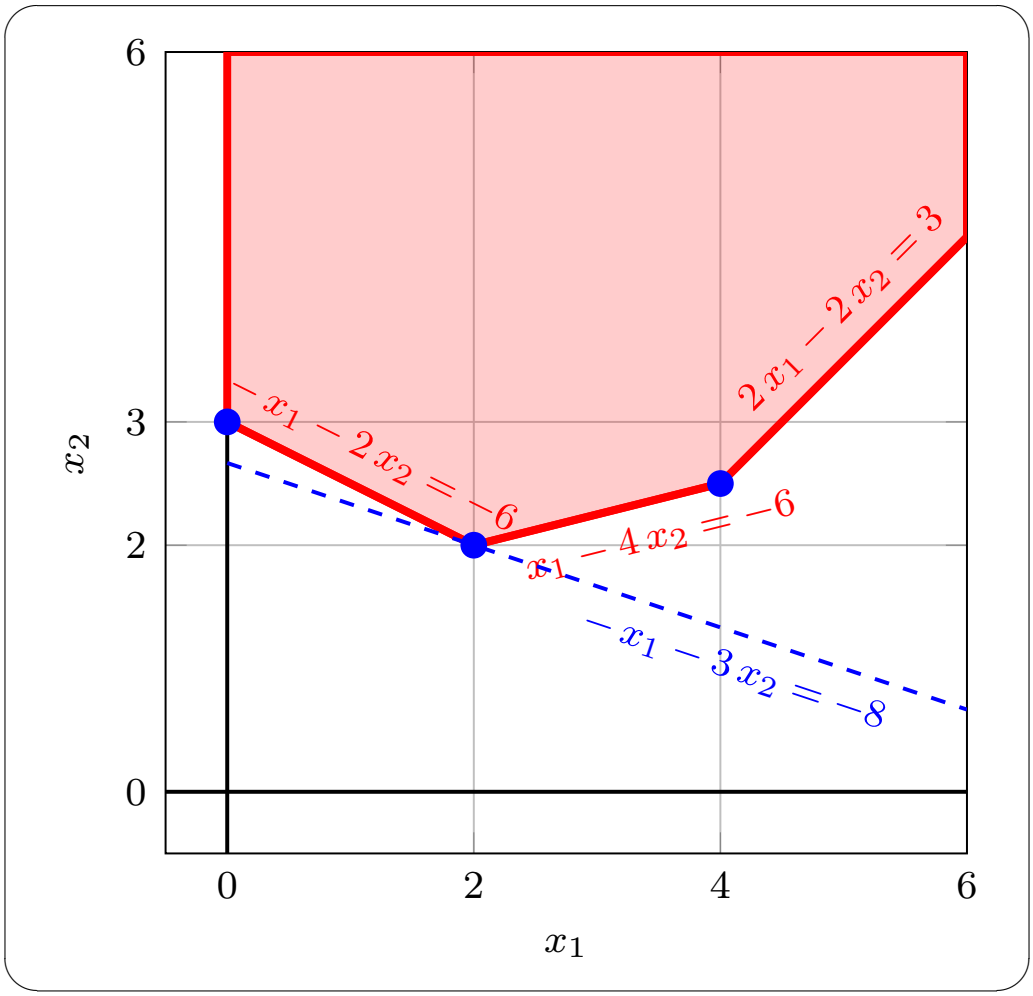
2.2 Problema LP *P12*

$$
\begin{array}{rrrrrr}
\max & -x_1 & - & 2\,x_2 & & \\
\text{soggetto a} & -x_1 & - & x_2 & \leq & -3 \\
& x_1 & - & 2\,x_2 & \leq & -1 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



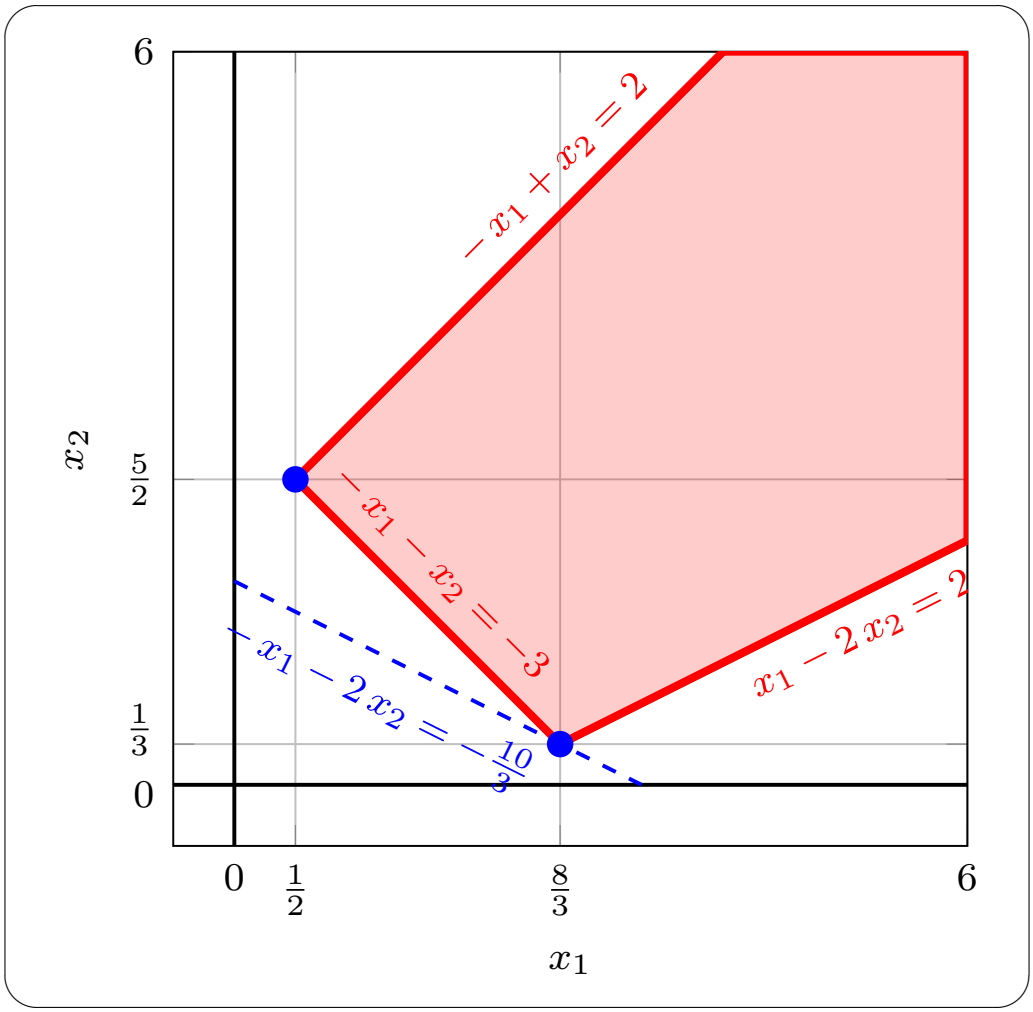
2.3 Problema LP *P*13

$$
\begin{array}{rrrrrrrr}
\max & - & x_1 & - & 3 & x_2 & & \\
\text{soggetto a} & - & x_1 & - & 2 & x_2 & \leq & -6 \\
 & & x_1 & - & 4 & x_2 & \leq & -6 \\
 & 2 & x_1 & - & 2 & x_2 & \leq & 3 \\
 & & x_1, & & & x_2 & \geq & 0
\end{array}
$$



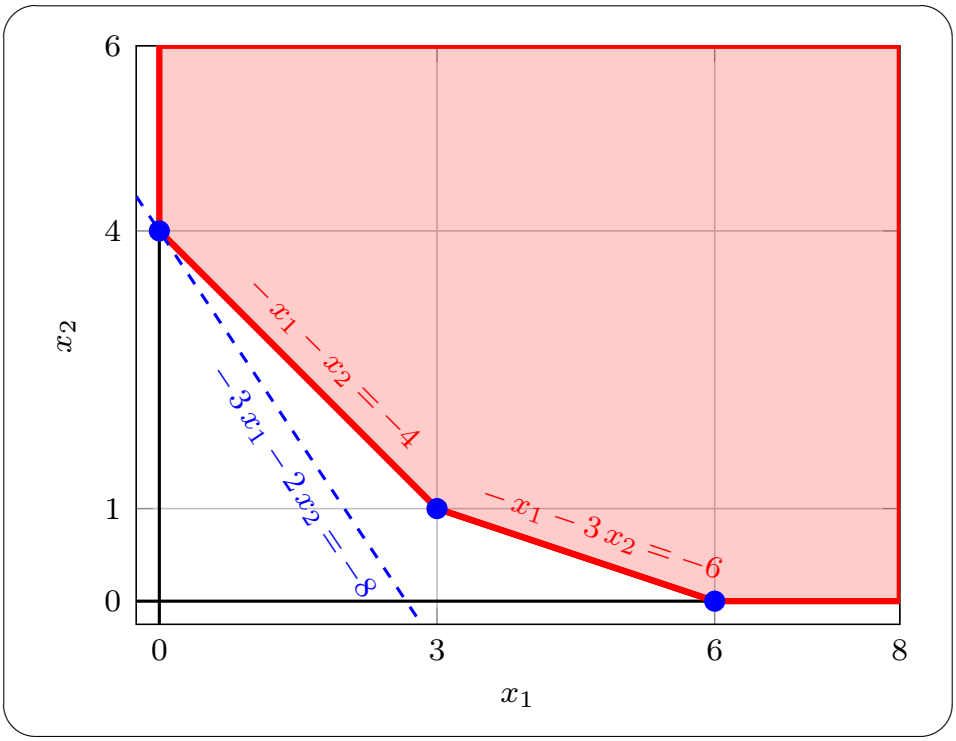
2.4 Problema LP *P14*

$$
\begin{array}{rrrrrrrr}
\max & - & x_1 & - & 2 & x_2 & & \\
\text{soggetto a} & - & x_1 & + & & x_2 & \leq & 2 \\
& & x_1 & - & 2 & x_2 & \leq & 2 \\
& - & x_1 & - & & x_2 & \leq & -3 \\
& & x_1, & & & x_2 & \geq & 0
\end{array}
$$



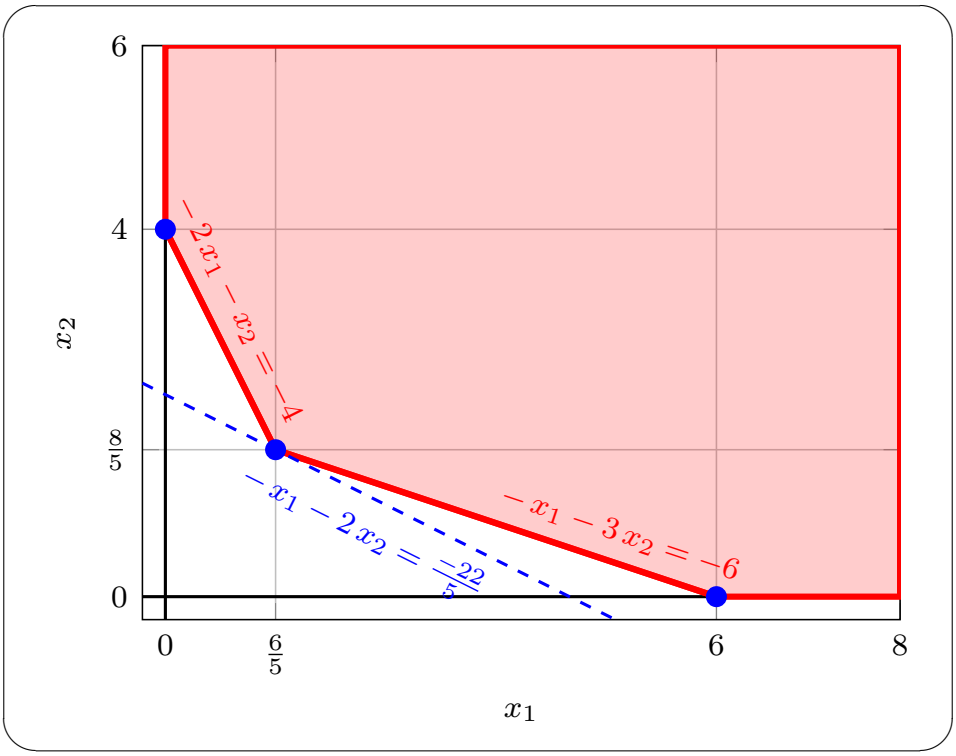
2.5 Problema LP *P*15

$$
\begin{array}{lrcrcr}
\max & -3\,x_1 & - & 2\,x_2 & & \\
\text{soggetto a} & -\,x_1 & - & x_2 & \leq & -4 \\
& -\,x_1 & - & 3\,x_2 & \leq & -6 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



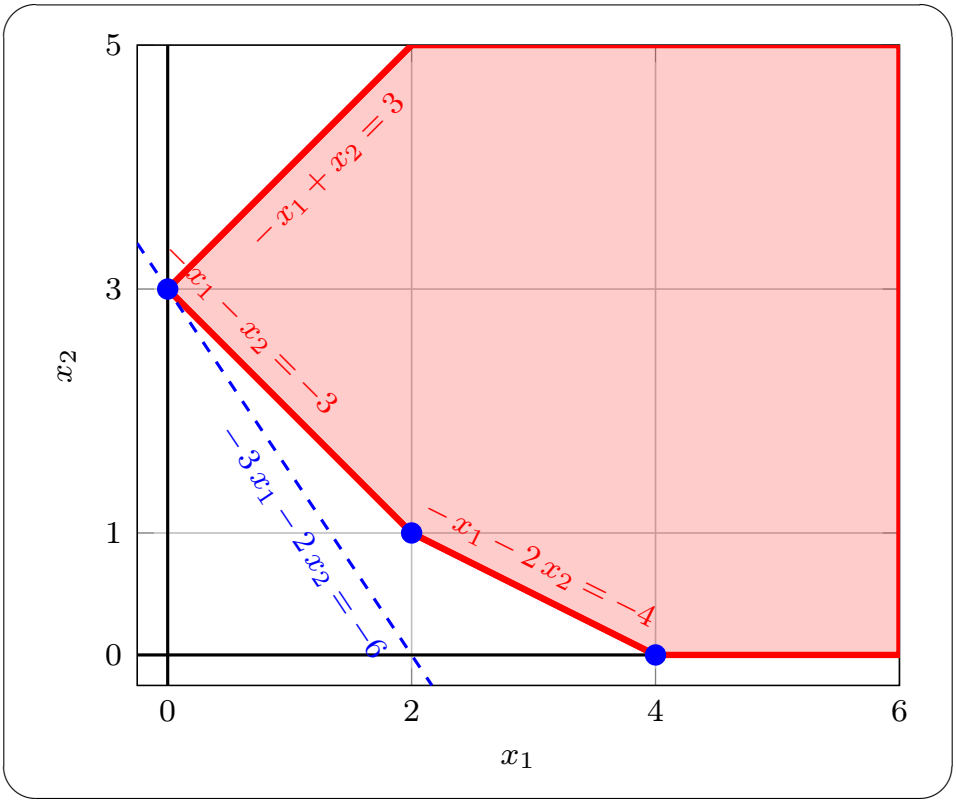
2.6 Problema LP *P16*

$$
\begin{array}{lrcrcr}
\max & -x_1 & - & 2x_2 & & \\
\text{soggetto a} & -2x_1 & - & x_2 & \leq & -4 \\
& -x_1 & - & 3x_2 & \leq & -6 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



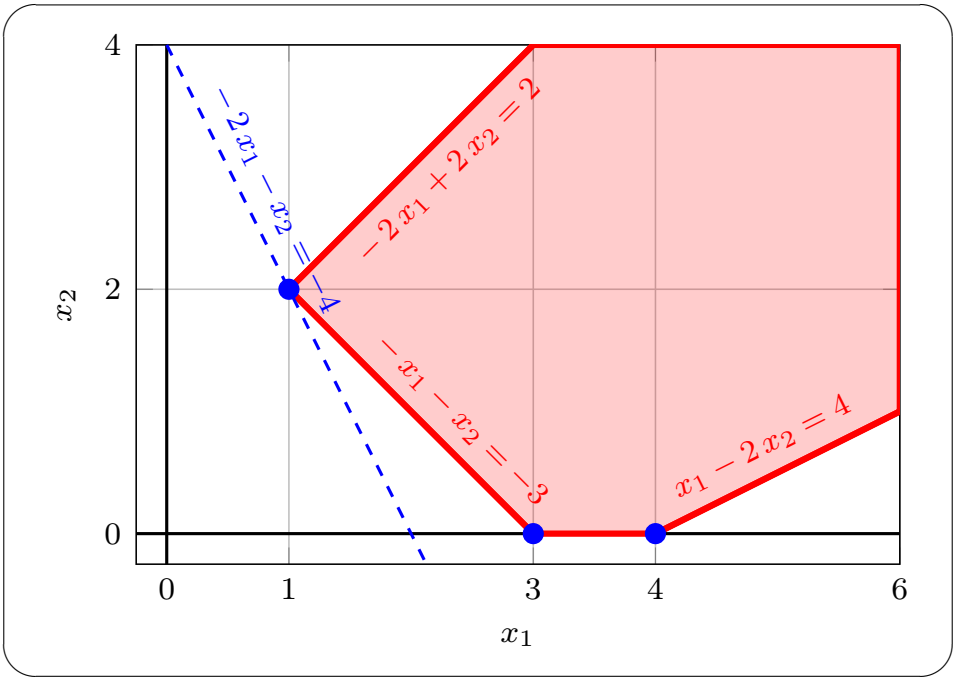
2.7 Problema LP *P17*

$$
\begin{array}{rrrrrr}
\max & -3\,x_1 & - & 2\,x_2 & & \\
\text{soggetto a} & -\,x_1 & - & x_2 & \leq & -3 \\
& -\,x_1 & + & x_2 & \leq & 3 \\
& -\,x_1 & - & 2\,x_2 & \leq & -4 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



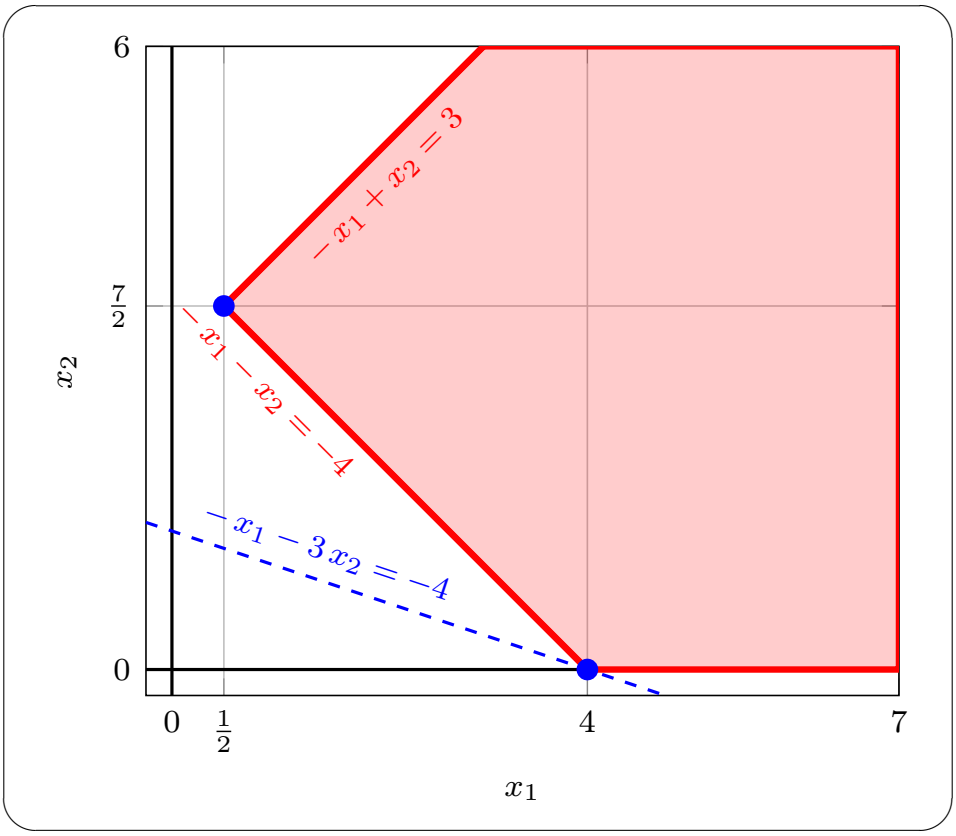
2.8 Problema LP *P18*

$$
\begin{array}{lrcrcr}
\max & -2\,x_1 & - & x_2 & & \\
\text{soggetto a} & -\,x_1 & - & x_2 & \leq & -3 \\
& x_1 & - & 2\,x_2 & \leq & 4 \\
& -2\,x_1 & + & 2\,x_2 & \leq & 2 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



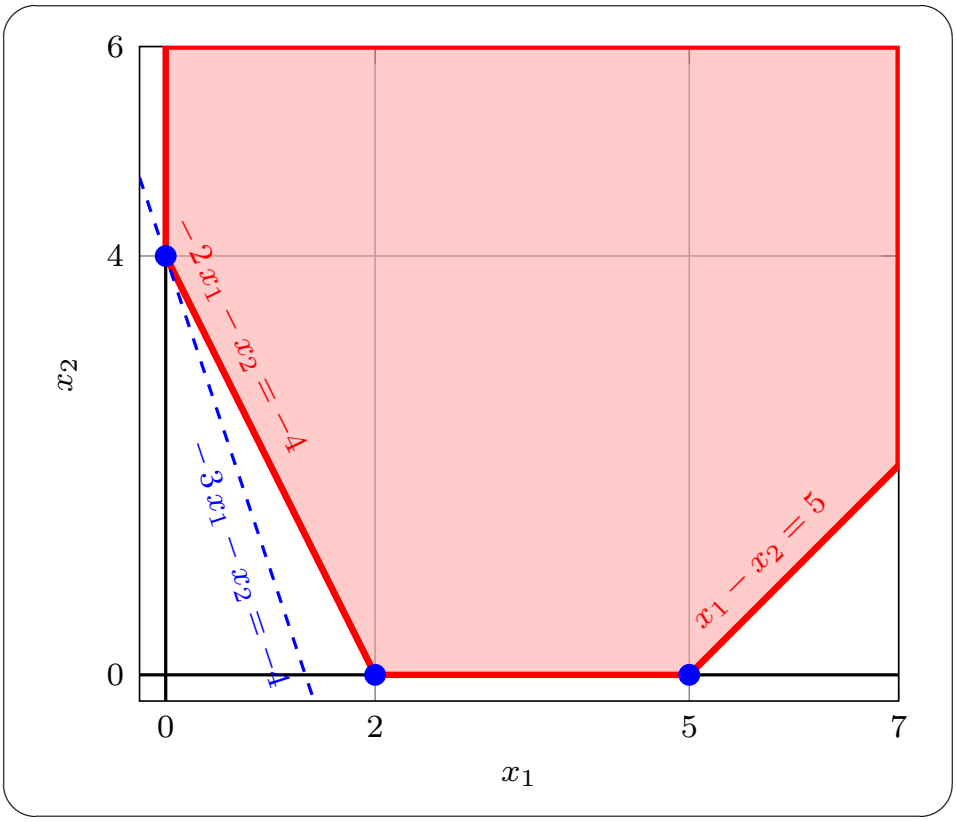
2.9 Problema LP *P19*

$$
\begin{array}{rrrrrr}
\max & -x_1 & - & 3\,x_2 & & \\
\text{soggetto a} & -x_1 & - & x_2 & \leq & -4 \\
& -x_1 & + & x_2 & \leq & 3 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



2.10 Problema LP *P20*

$$
\begin{array}{rrcrcr}
\max & -3\,x_1 & - & x_2 & & \\
\text{soggetto a} & x_1 & - & x_2 & \leq & 5 \\
& -2\,x_1 & - & x_2 & \leq & -4 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$

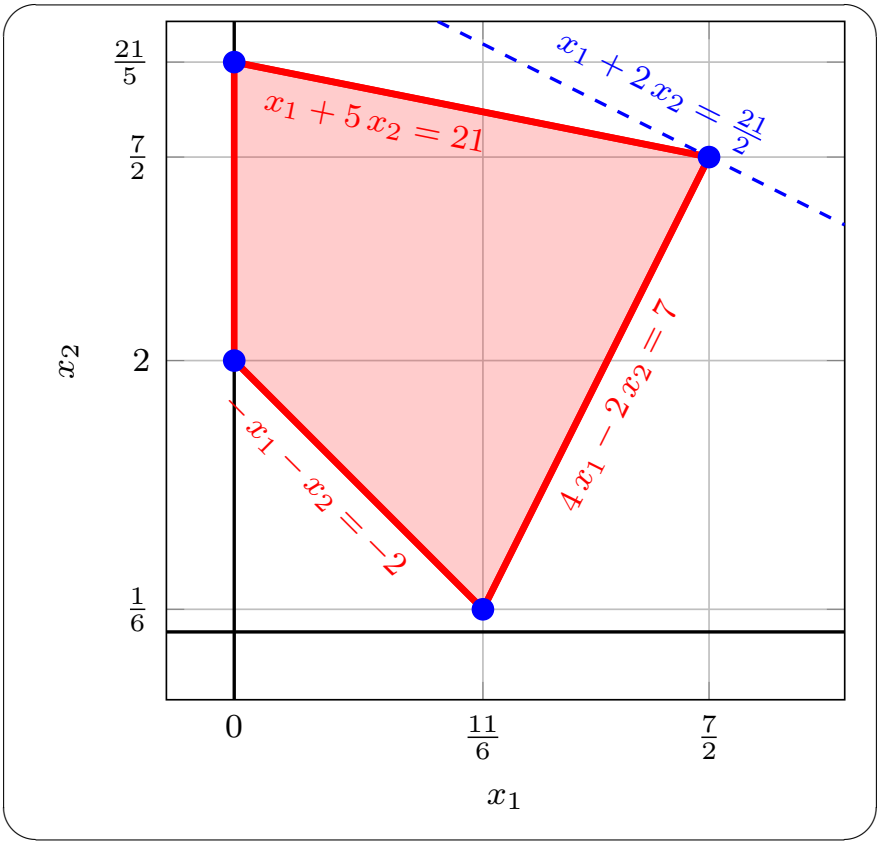


Capitolo 3

Problemi LP con due variabili – Tipo C

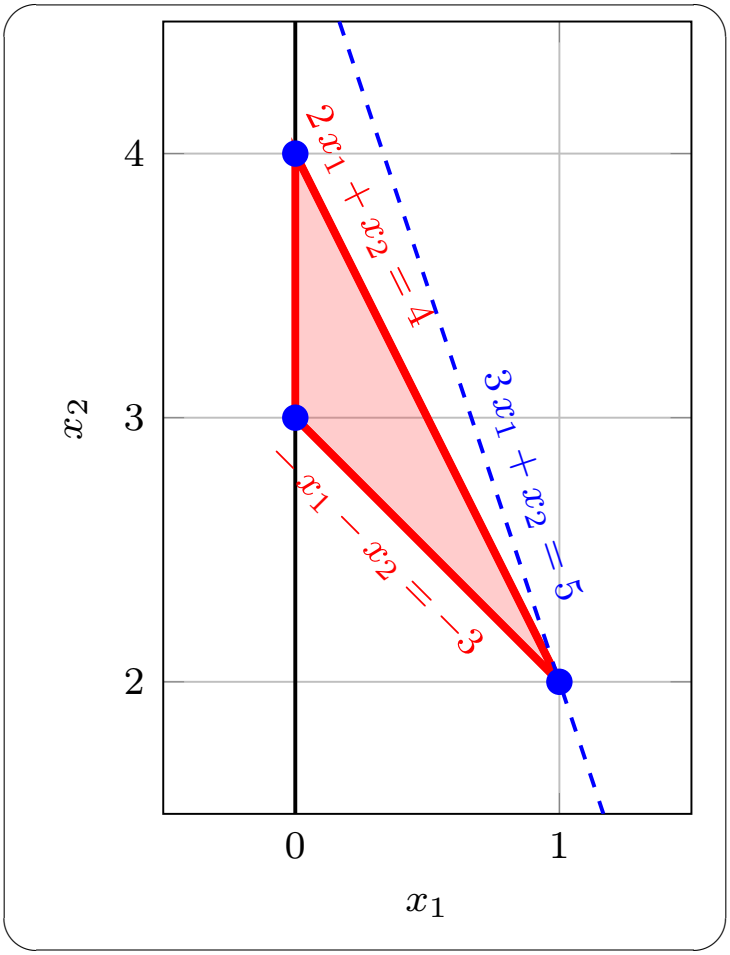
3.1 Problema LP *P21*

$$
\begin{array}{rrrrrr}
\max & x_1 & + & 2\,x_2 & & \\
\text{soggetto a} & -\,x_1 & - & x_2 & \leq & -2 \\
& 4\,x_1 & - & 2\,x_2 & \leq & 7 \\
& x_1 & + & 5\,x_2 & \leq & 21 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



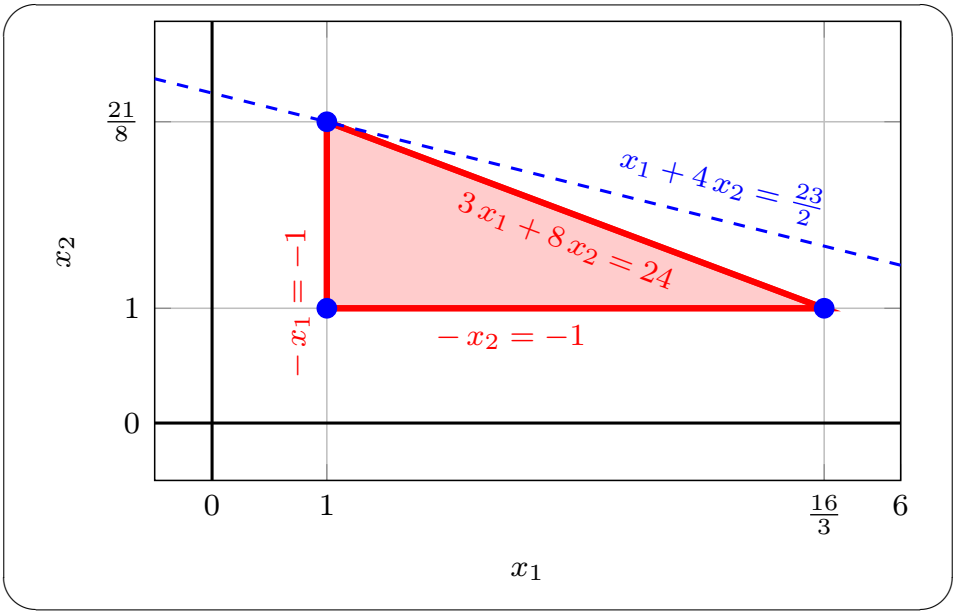
3.2 Problema LP *P22*

$$
\begin{array}{lrcrcr}
\max & 3\,x_1 & + & x_2 & & \\
\text{soggetto a} & x_1 & - & x_2 & \leq & -1 \\
& -\,x_1 & - & x_2 & \leq & -3 \\
& 2\,x_1 & + & x_2 & \leq & 4 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



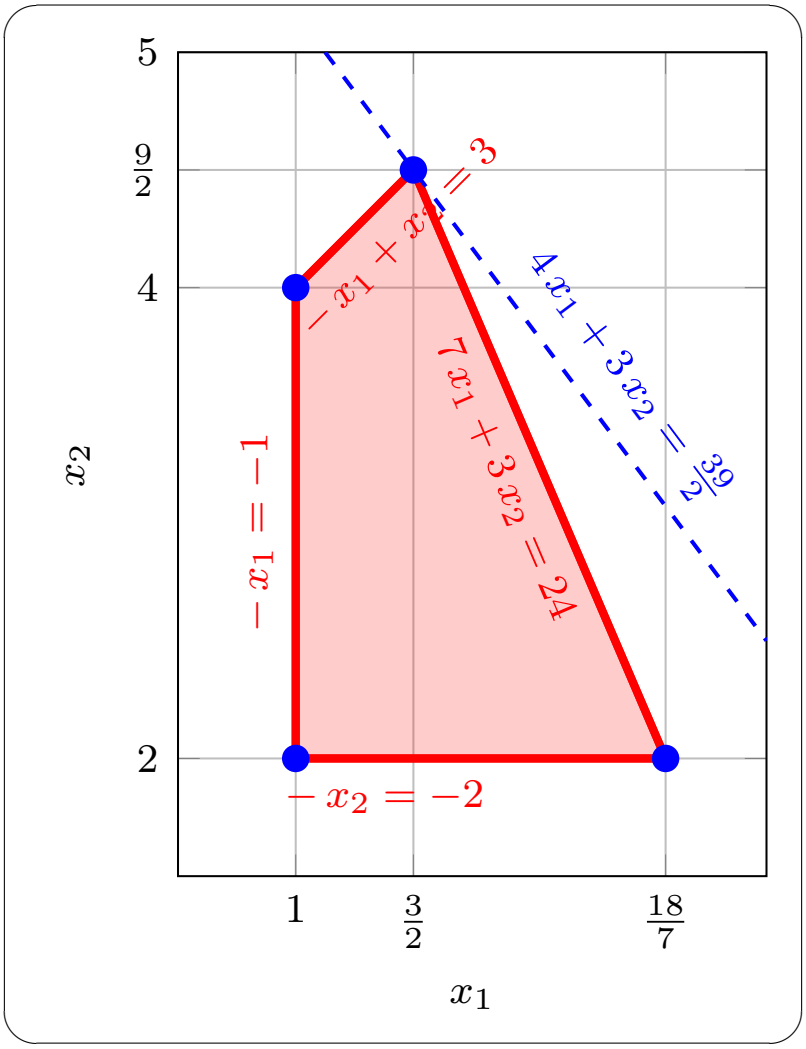
3.3 Problema LP *P23*

$$
\begin{array}{lrcrcl}
\max & x_1 & + & 4\,x_2 & & \\
\text{soggetto a} & 3\,x_1 & + & 8\,x_2 & \leq & 24 \\
 & x_1 & & & \geq & 1 \\
 & & & x_2 & \geq & 1 \\
 & x_1, & & x_2 & \geq & 0
\end{array}
$$



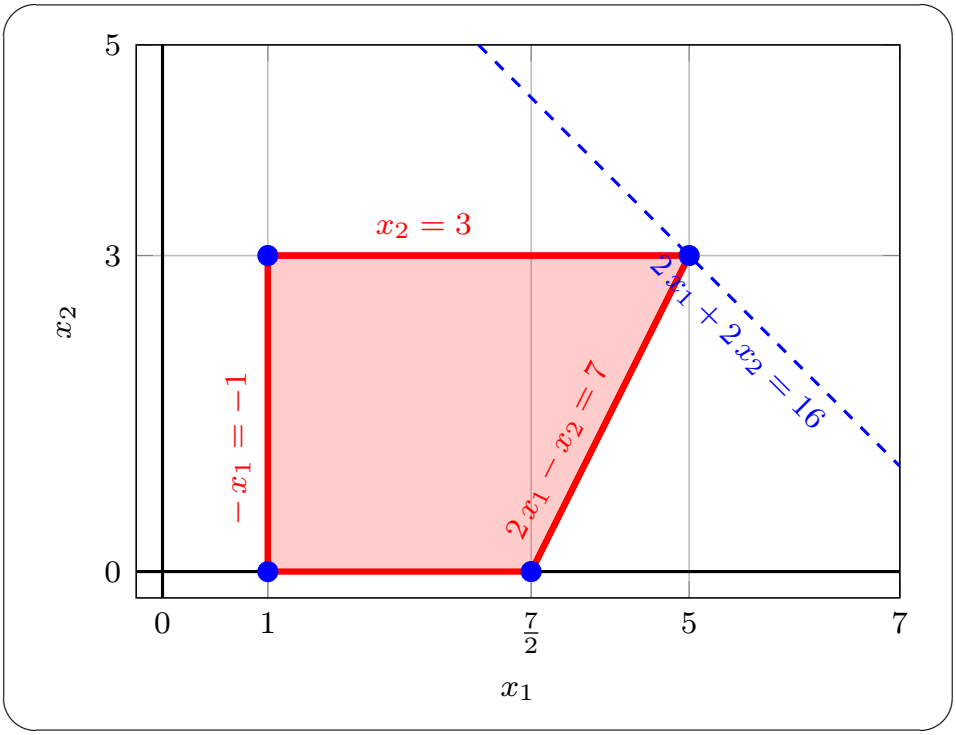
3.4 Problema LP *P24*

$$
\begin{array}{rrcrcr}
\max & 4\,x_1 & + & 3\,x_2 & & \\
\text{soggetto a} & 7\,x_1 & + & 3\,x_2 & \leq & 24 \\
 & -\,x_1 & + & x_2 & \leq & 3 \\
 & x_1 & & & \geq & 1 \\
 & & & x_2 & \geq & 2 \\
 & x_1, & & x_2 & \geq & 0
\end{array}
$$



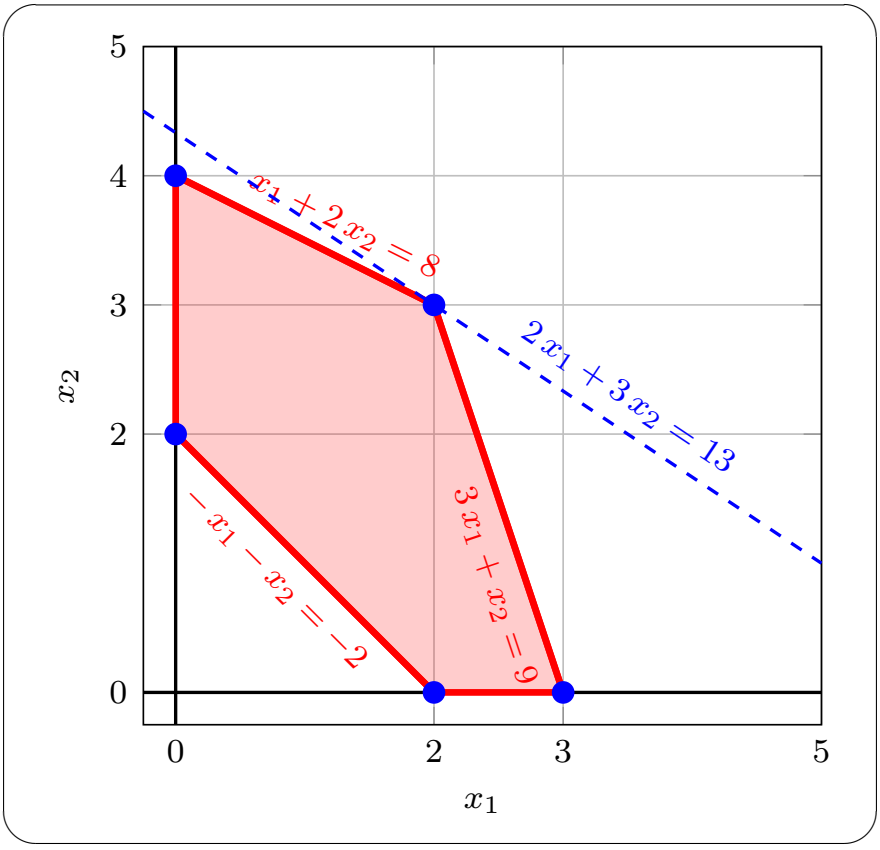
3.5 Problema LP *P25*

$$
\begin{array}{rrcrcl}
\max & 2\,x_1 & + & 2\,x_2 & & \\
\text{soggetto a} & x_1 & & & \geq & 1 \\
 & & & x_2 & \leq & 3 \\
 & 2\,x_1 & - & x_2 & \leq & 7 \\
 & x_1, & & x_2 & \geq & 0
\end{array}
$$



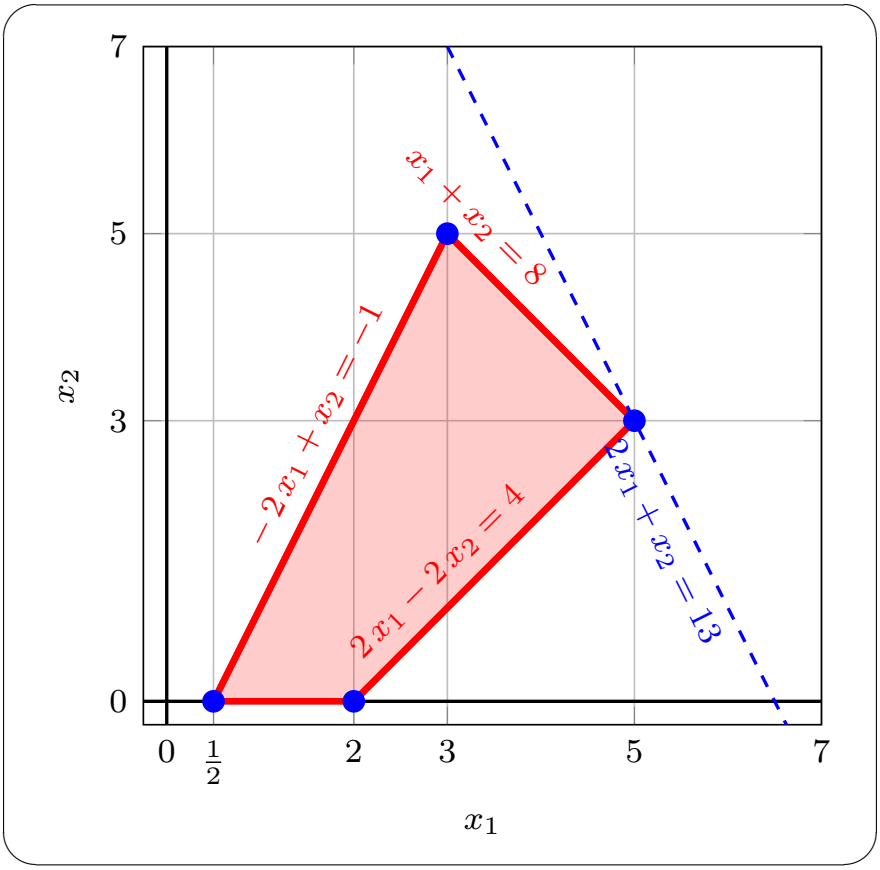
3.6 Problema LP *P26*

$$
\begin{array}{rrcrcr}
\max & 2\,x_1 & + & 3\,x_2 & & \\
\text{soggetto a} & -\,x_1 & - & x_2 & \leq & -2 \\
& x_1 & + & 2\,x_2 & \leq & 8 \\
& 3\,x_1 & + & x_2 & \leq & 9 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



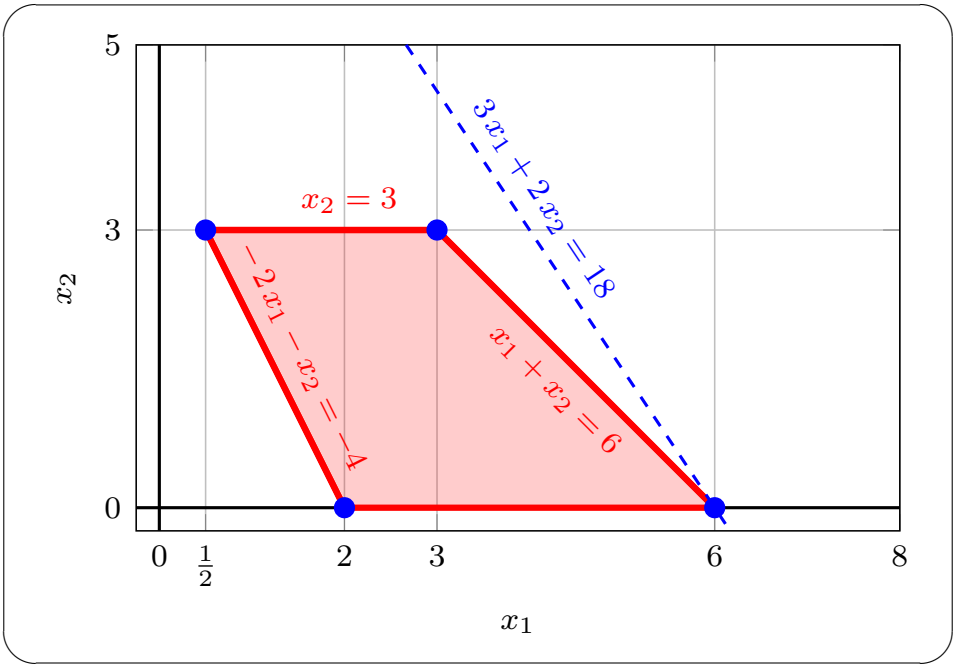
3.7 Problema LP *P27*

$$
\begin{array}{rrcrcr}
\max & 2\,x_1 & + & x_2 & & \\
\text{soggetto a} & x_1 & + & x_2 & \leq & 8 \\
& 2\,x_1 & - & 2\,x_2 & \leq & 4 \\
& -2\,x_1 & + & x_2 & \leq & -1 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



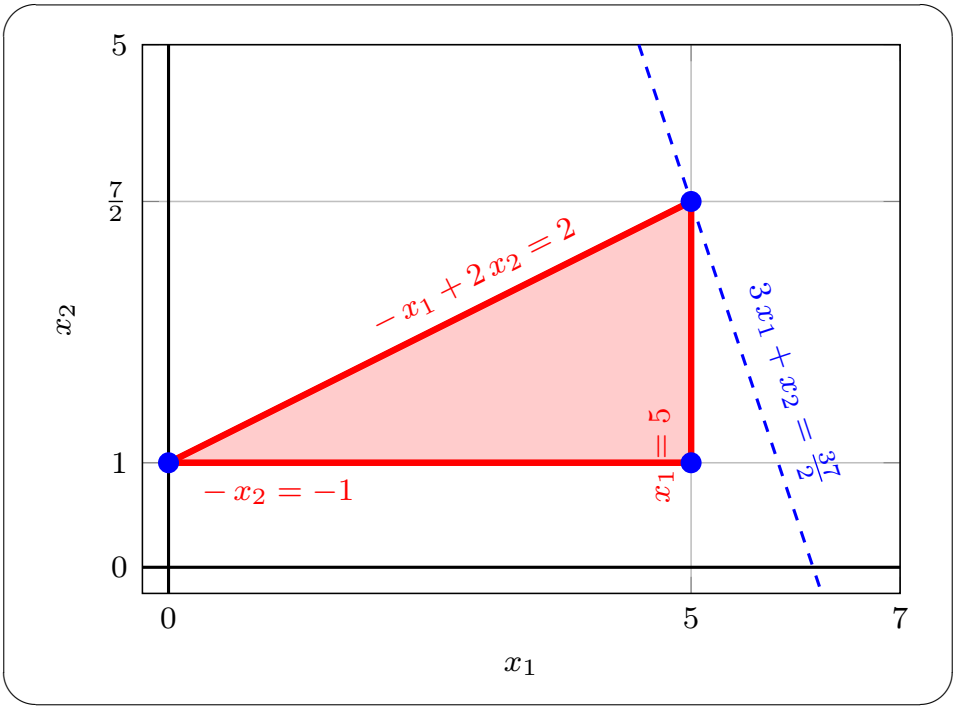
3.8 Problema LP *P28*

$$
\begin{array}{lrcrcr}
\max & 3\,x_1 & + & 2\,x_2 & & \\
\text{soggetto a} & x_1 & + & x_2 & \leq & 6 \\
& -2\,x_1 & - & x_2 & \leq & -4 \\
& & & x_2 & \leq & 3 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



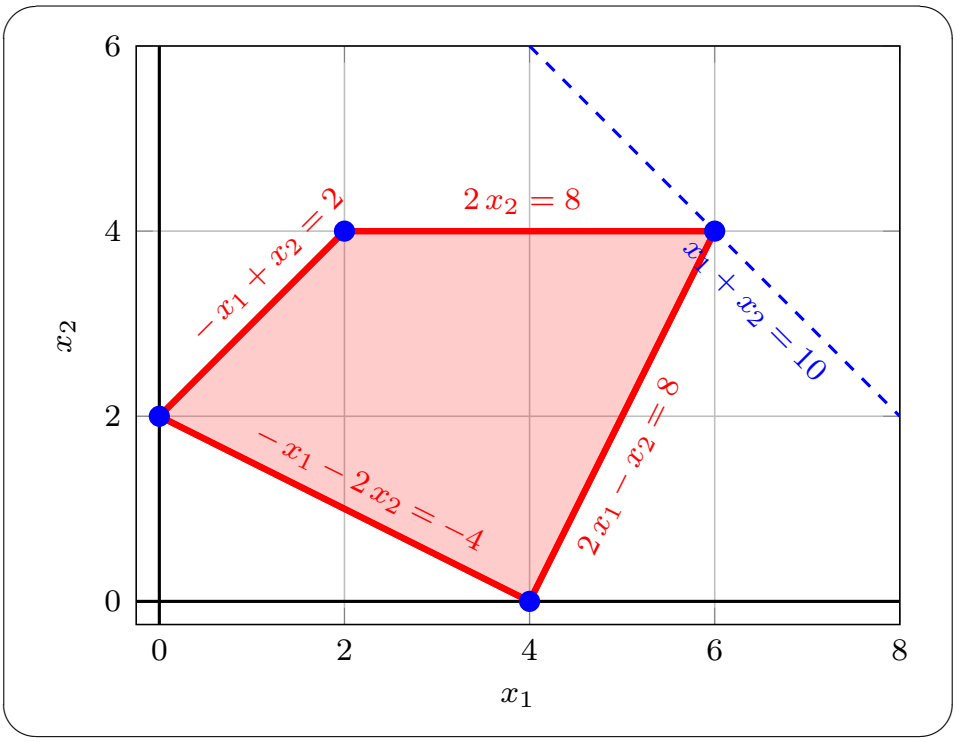
3.9 Problema LP *P29*

$$
\begin{array}{rrcrcl}
\max & 3\,x_1 & + & x_2 & & \\
\text{soggetto a} & -\,x_1 & + & 2\,x_2 & \leq & 2 \\
 & & & x_2 & \geq & 1 \\
 & x_1 & & & \leq & 5 \\
 & x_1, & & x_2 & \geq & 0
\end{array}
$$



3.10 Problema LP *P30*

$$
\begin{array}{rrrrrr}
\max & x_1 & + & x_2 & & \\
\text{soggetto a} & -x_1 & + & x_2 & \leq & 2 \\
& -x_1 & - & 2\,x_2 & \leq & -4 \\
& 2\,x_1 & - & x_2 & \leq & 8 \\
& & & 2\,x_2 & \leq & 8 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$

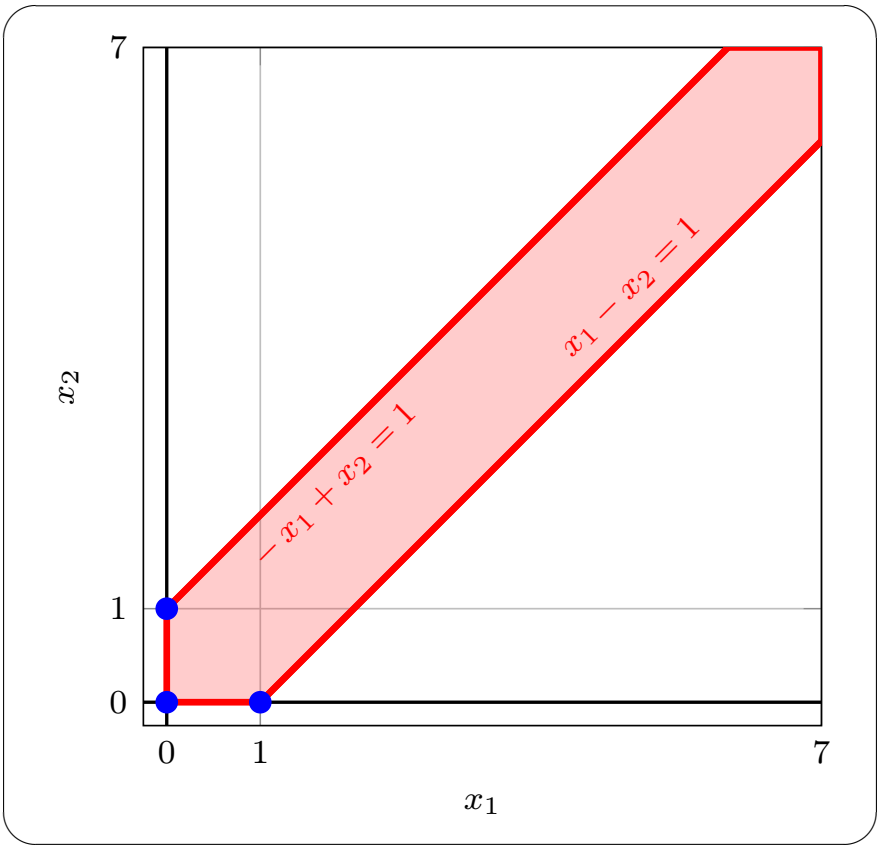


Capitolo 4

Problemi LP con due variabili – Tipo D

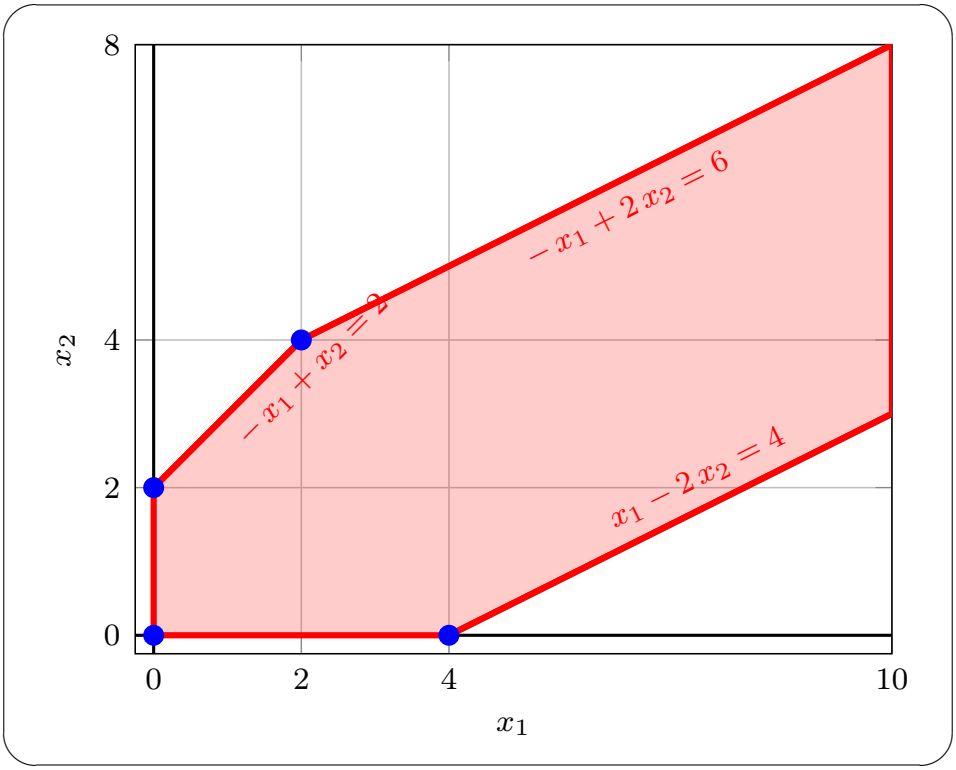
4.1 Problema LP *P31*

$$
\begin{array}{rrcrcl}
\max & x_1 & + & x_2 & & \\
\text{soggetto a} & -x_1 & + & x_2 & \leq & 1 \\
& x_1 & - & x_2 & \leq & 1 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



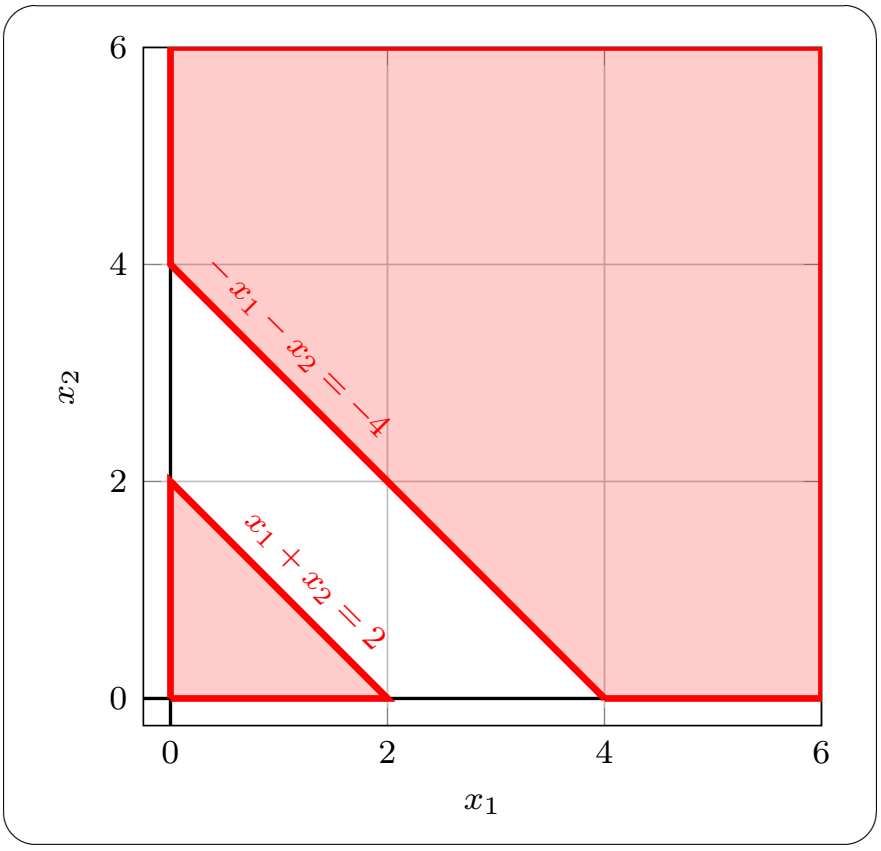
4.2 Problema LP *P32*

$$
\begin{array}{rrcrcl}
\max & 2\,x_1 & + & x_2 & & \\
\text{soggetto a} & -\,x_1 & + & x_2 & \leq & 2 \\
& -\,x_1 & + & 2\,x_2 & \leq & 6 \\
& x_1 & - & 2\,x_2 & \leq & 4 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



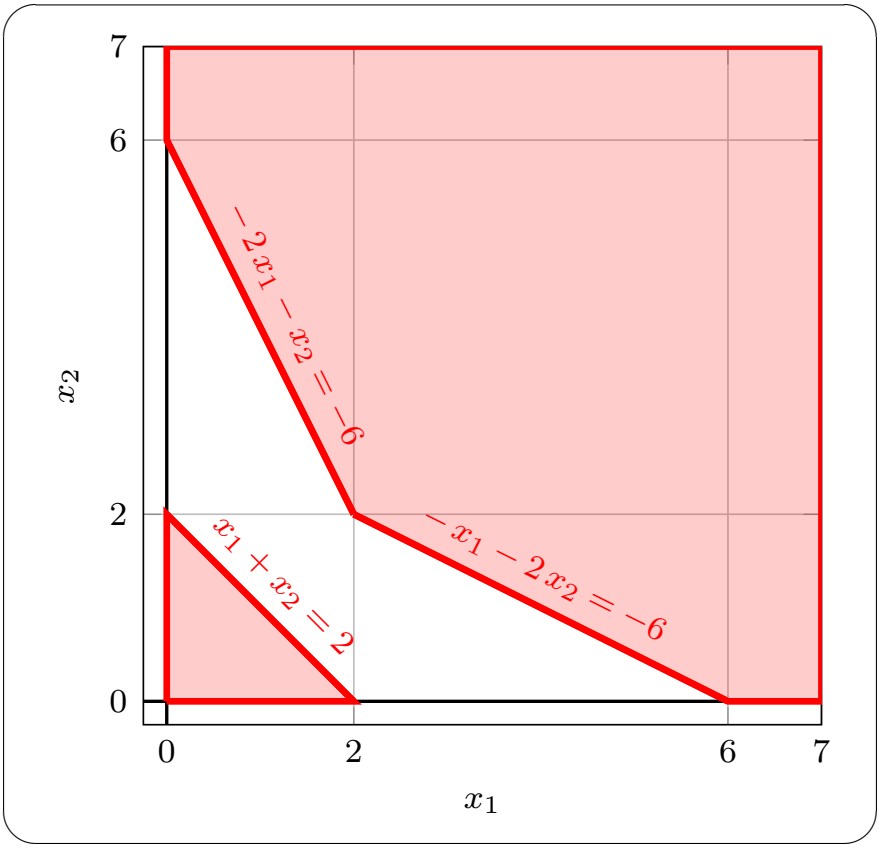
4.3 Problema LP *P33*

$$
\begin{array}{lrcrcr}
\max & x_1 & + & x_2 & & \\
\text{soggetto a} & x_1 & + & x_2 & \leq & 2 \\
& -x_1 & - & x_2 & \leq & -4 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



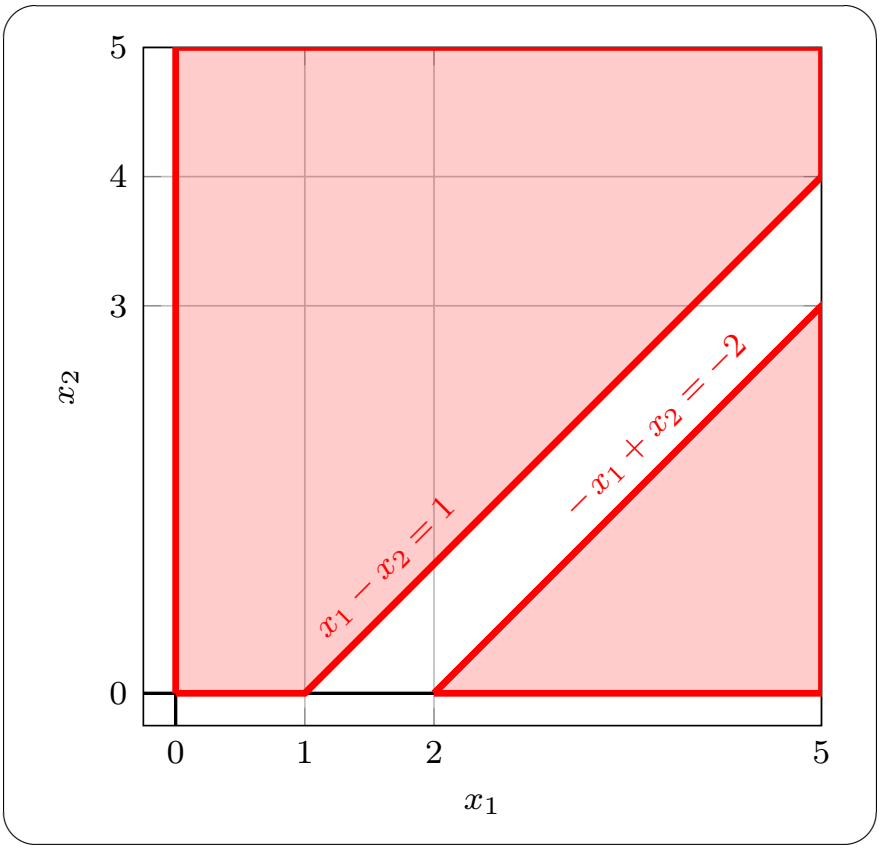
4.4 Problema LP *P34*

$$
\begin{array}{rrrrrr}
\max & -2\,x_1 & - & x_2 & & \\
\text{soggetto a} & -\,x_1 & - & 2\,x_2 & \leq & -6 \\
& -2\,x_1 & - & x_2 & \leq & -6 \\
& x_1 & + & x_2 & \leq & 2 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



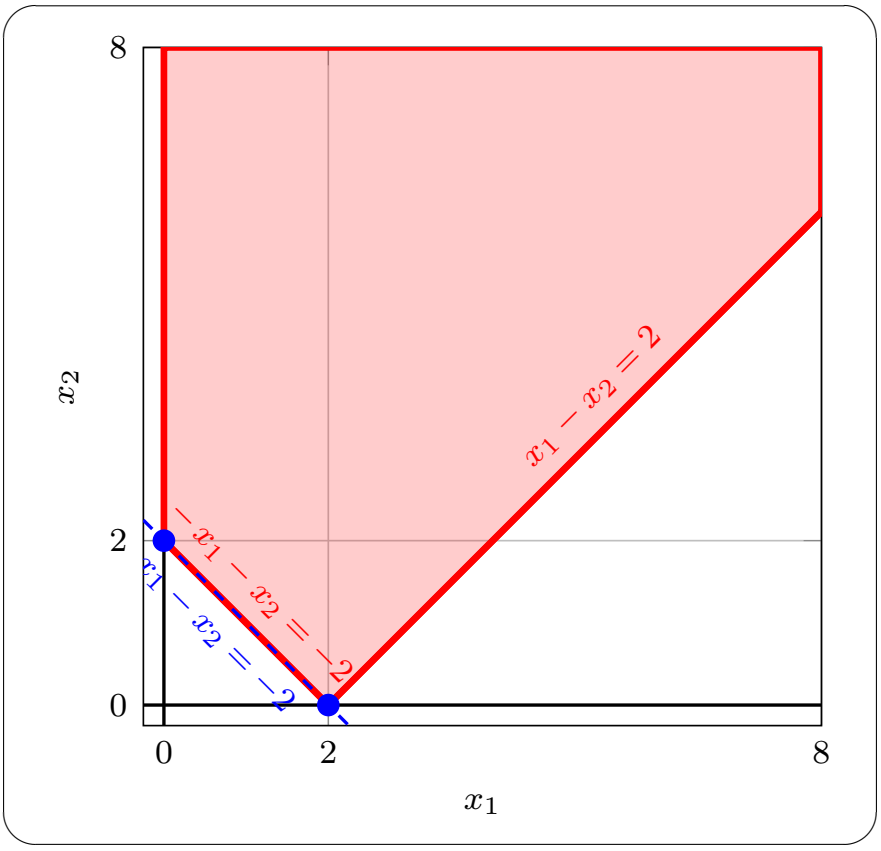
4.5 Problema LP *P35*

$$
\begin{array}{lrcrcr}
\max & x_1 & + & x_2 & & \\
\text{soggetto a} & x_1 & - & x_2 & \leq & 1 \\
& -x_1 & + & x_2 & \leq & -2 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



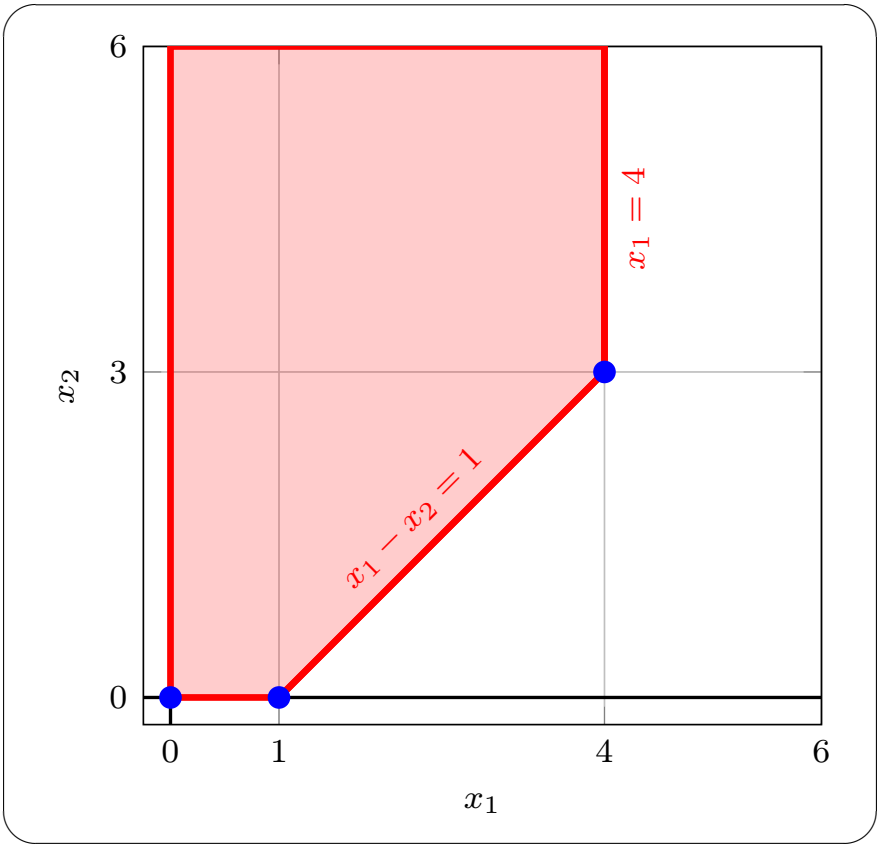
4.6 Problema LP *P36*

$$
\begin{array}{lrcrcr}
\max & -x_1 & - & x_2 & & \\
\text{soggetto a} & -x_1 & - & x_2 & \leq & -2 \\
& x_1 & - & x_2 & \leq & 2 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



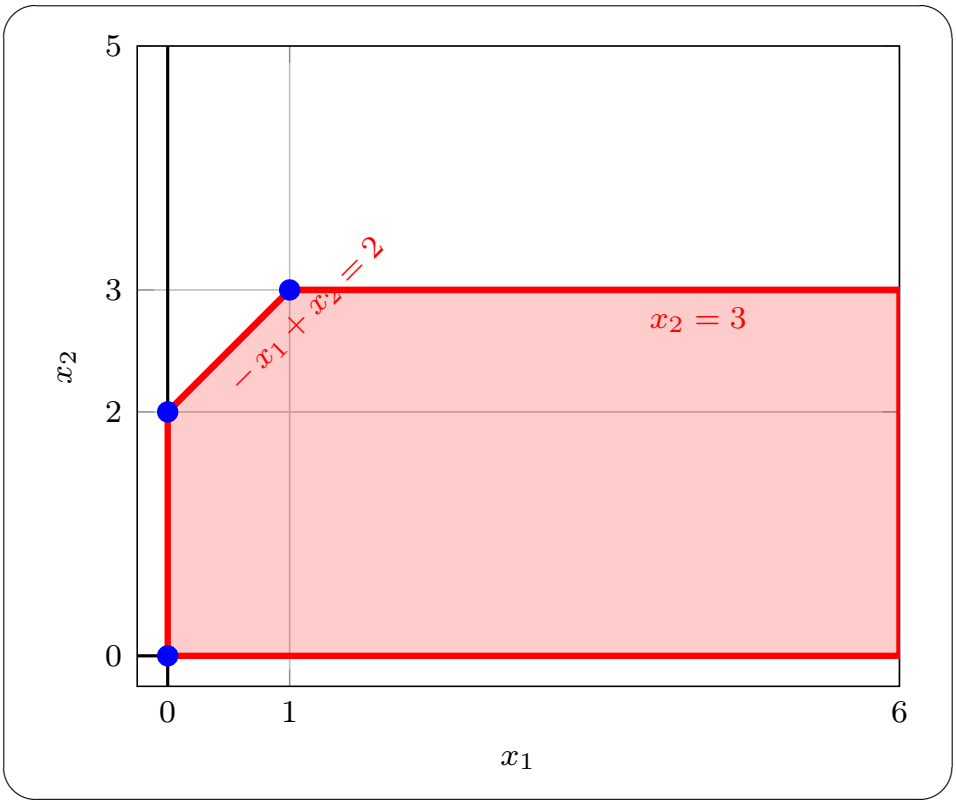
4.7 Problema LP *P37*

$$
\begin{array}{rrcrcl}
\max & x_1 & + & 2\,x_2 & & \\
\text{soggetto a} & x_1 & - & x_2 & \leq & 1 \\
 & x_1 & & & \leq & 4 \\
 & x_1, & & x_2 & \geq & 0
\end{array}
$$



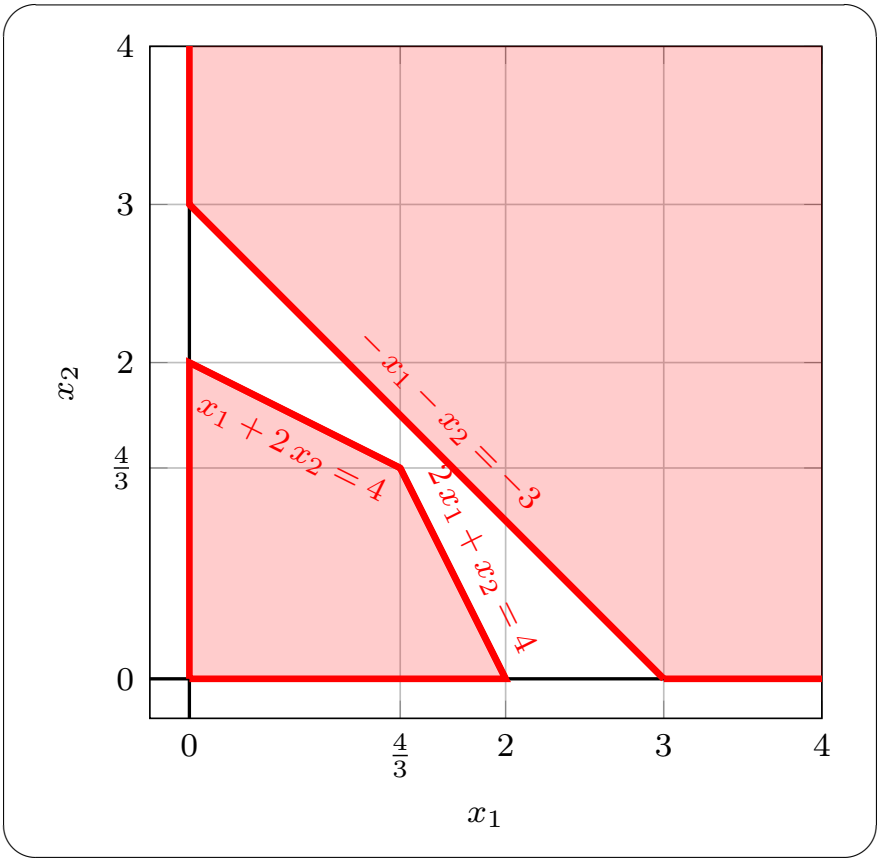
4.8 Problema LP *P38*

$$
\begin{array}{rrcrcl}
\max & 2\,x_1 & + & x_2 & & \\
\text{soggetto a} & -\,x_1 & + & x_2 & \leq & 2 \\
 & & & x_2 & \leq & 3 \\
 & x_1, & & x_2 & \geq & 0
\end{array}
$$



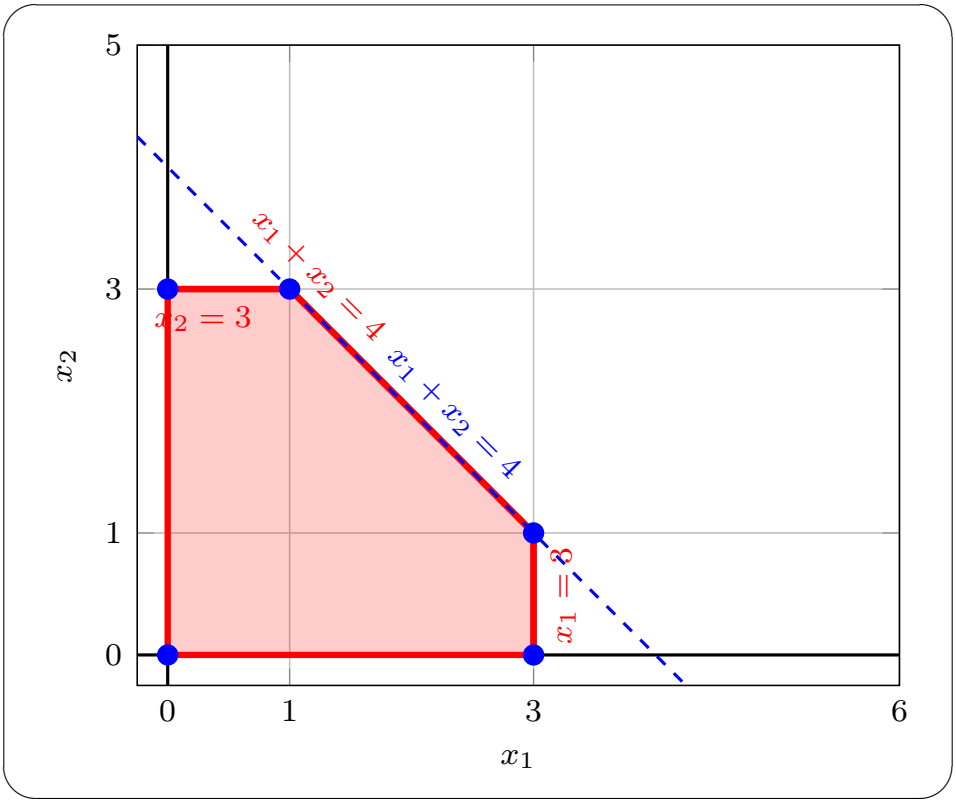
4.9 Problema LP *P39*

$$
\begin{array}{rrcrcr}
\max & x_1 & + & x_2 & & \\
\text{soggetto a} & 2\,x_1 & + & x_2 & \leq & 4 \\
& x_1 & + & 2\,x_2 & \leq & 4 \\
& -\,x_1 & - & x_2 & \leq & -3 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



4.10 Problema LP *P40*

$$
\begin{array}{rrcrcl}
\max & x_1 & + & x_2 & & \\
\text{soggetto a} & x_1 & + & x_2 & \leq & 4 \\
 & x_1 & & & \leq & 3 \\
 & & & x_2 & \leq & 3 \\
 & x_1, & & x_2 & \geq & 0
\end{array}
$$

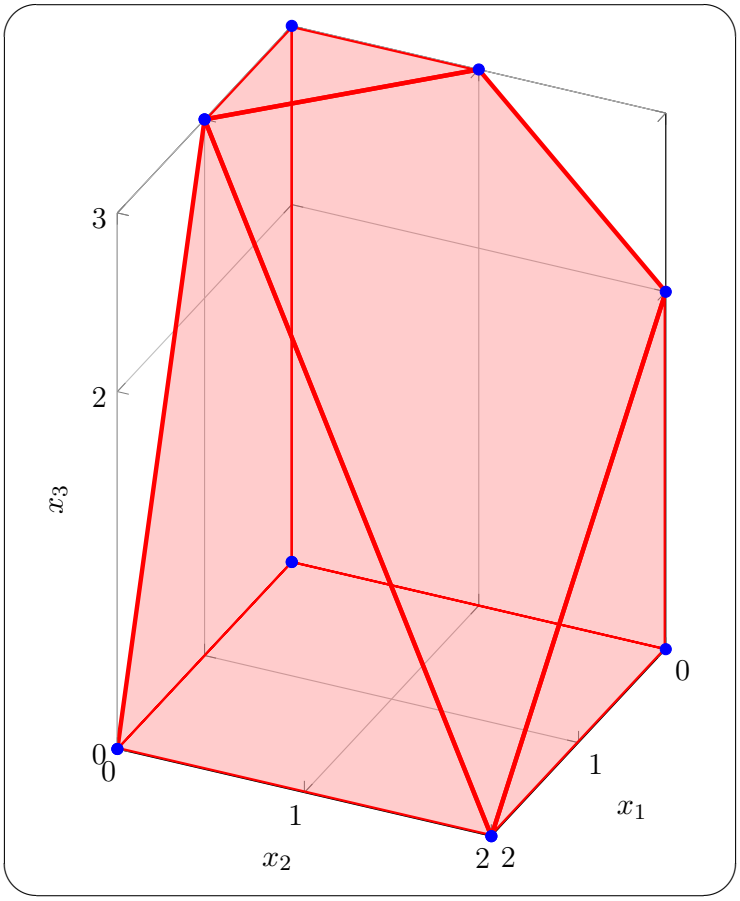


Capitolo 5

Problemi LP con tre variabili – Tipo A

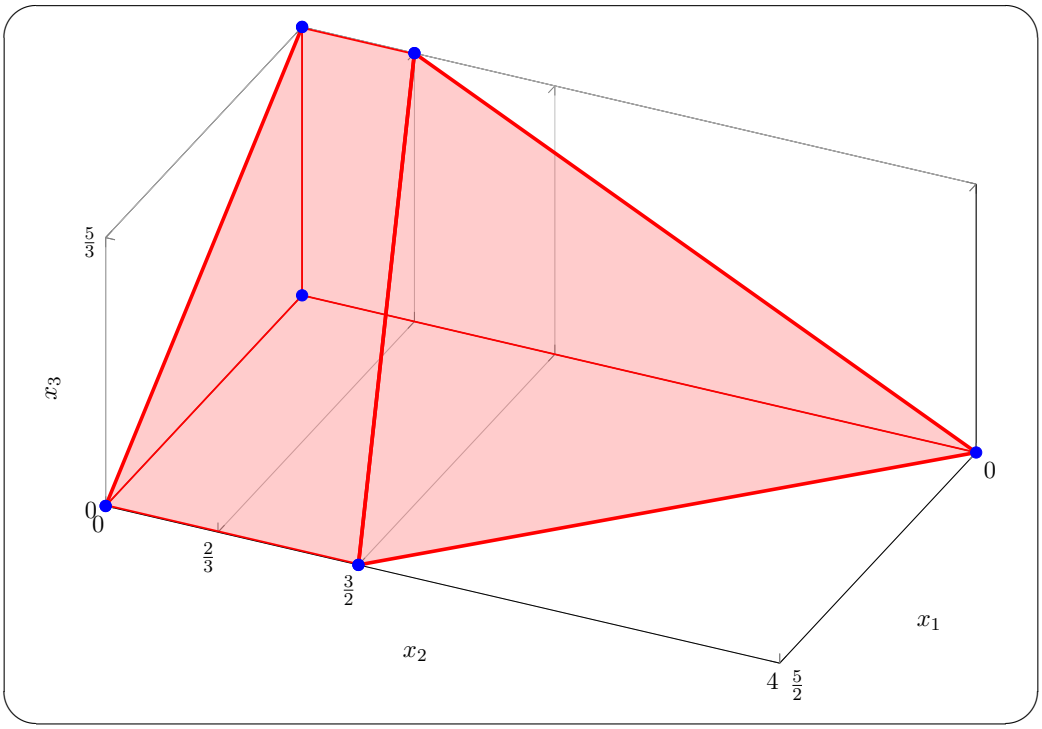
5.1 Problema LP *P41*

$$
\begin{array}{rrcrcrcl}
\max & 2\,x_1 & + & x_2 & + & 3\,x_3 & & \\
\text{soggetto a} & x_1 & + & x_2 & + & x_3 & \leq & 4 \\
 & & & x_2 & & & \leq & 2 \\
 & & & & & x_3 & \leq & 3 \\
 & 3\,x_1 & & & + & x_3 & \leq & 6 \\
 & x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$



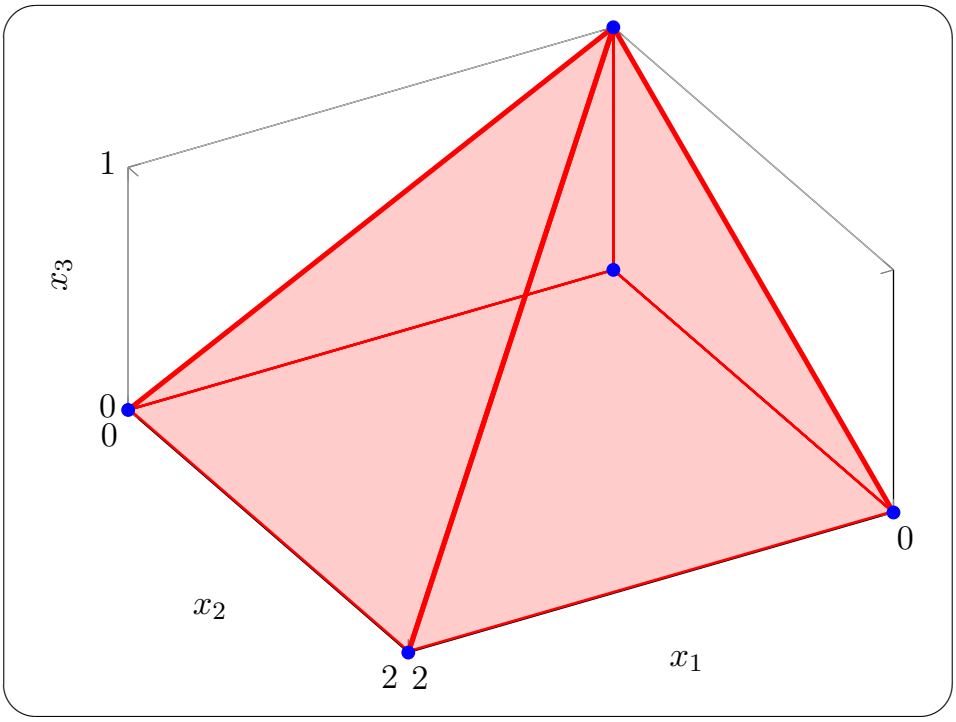
5.2 Problema LP *P42*

$$
\begin{array}{rrcrcrcrcl}
\max & 3 & x_1 & + & 2 & x_2 & + & 4 & x_3 & \\
\text{soggetto a} & & x_1 & + & & x_2 & + & 2 & x_3 & \leq & 4 \\
& 2 & x_1 & & & & + & 3 & x_3 & \leq & 5 \\
& 2 & x_1 & + & & x_2 & + & 3 & x_3 & \leq & 7 \\
& & x_1, & & & x_2, & & & x_3 & \geq & 0
\end{array}
$$



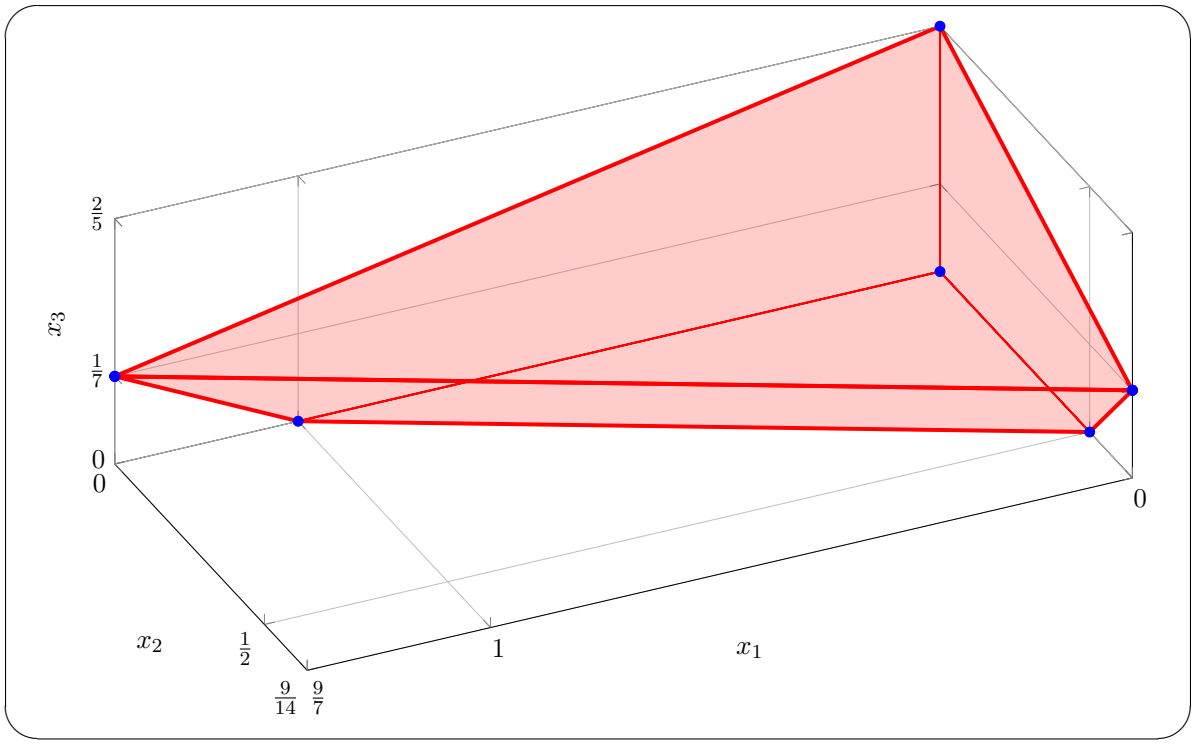
5.3 Problema LP *P43*

$$
\begin{array}{lrcrcrcrl}
\max & x_1 & + & 2 & x_2 & + & 7 & x_3 & \\
\text{soggetto a} & x_1 & & & & + & 2 & x_3 & \leq \quad 2 \\
& & & & x_2 & + & 2 & x_3 & \leq \quad 2 \\
& x_1, & & & x_2, & & & x_3 & \geq \quad 0
\end{array}
$$



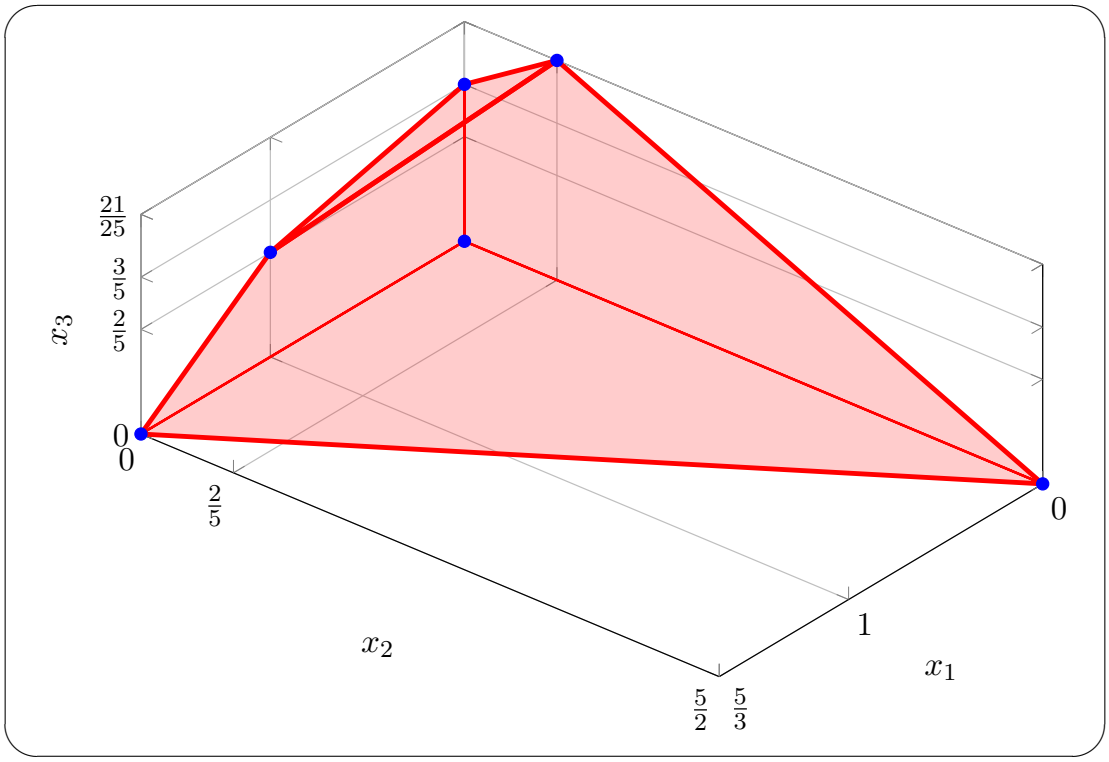
5.4 Problema LP *P44*

$$
\begin{array}{rrcrcrcrcl}
\max & x_1 & + & 2 & x_2 & + & 3 & x_3 & & \\
\text{soggetto a} & x_1 & + & 2 & x_2 & - & 2 & x_3 & \leq & 1 \\
2 & x_1 & - & & x_2 & - & 3 & x_3 & \leq & 4 \\
& x_1 & + & 2 & x_2 & + & 5 & x_3 & \leq & 2 \\
& x_1, & & & x_2, & & & x_3 & \geq & 0
\end{array}
$$



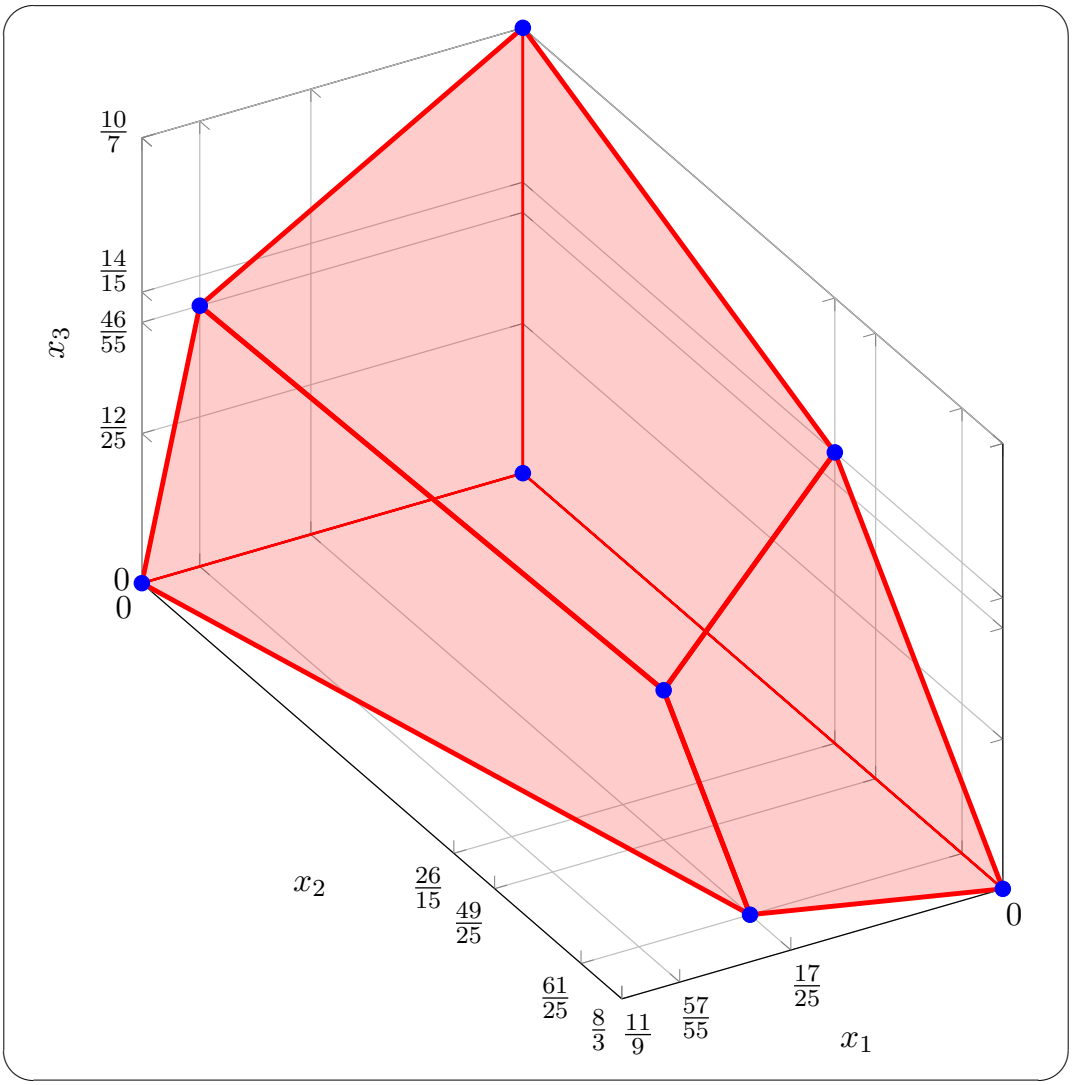
5.5 Problema LP *P45*

$$
\begin{array}{rrcrcrcl}
\max & 4\,x_1 & - & 13\,x_2 & + & 7\,x_3 & & \\
\text{soggetto a} & 3\,x_1 & + & 2\,x_2 & + & 5\,x_3 & \leq & 5 \\
& x_1 & - & 3\,x_2 & + & 5\,x_3 & \leq & 3 \\
& x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$



5.6 Problema LP *P46*

$$
\begin{array}{rrcrcrcr}
\max & 5\ x_1 & + & 6\ x_2 & + & 10\ x_3 & & \\
\text{soggetto a} & 9\ x_1 & + & 2\ x_2 & + & 2\ x_3 & \leq & 11 \\
& 2\ x_1 & + & 6\ x_2 & + & 6\ x_3 & \leq & 16 \\
& 4\ x_1 & + & 2\ x_2 & + & 7\ x_3 & \leq & 10 \\
& x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$

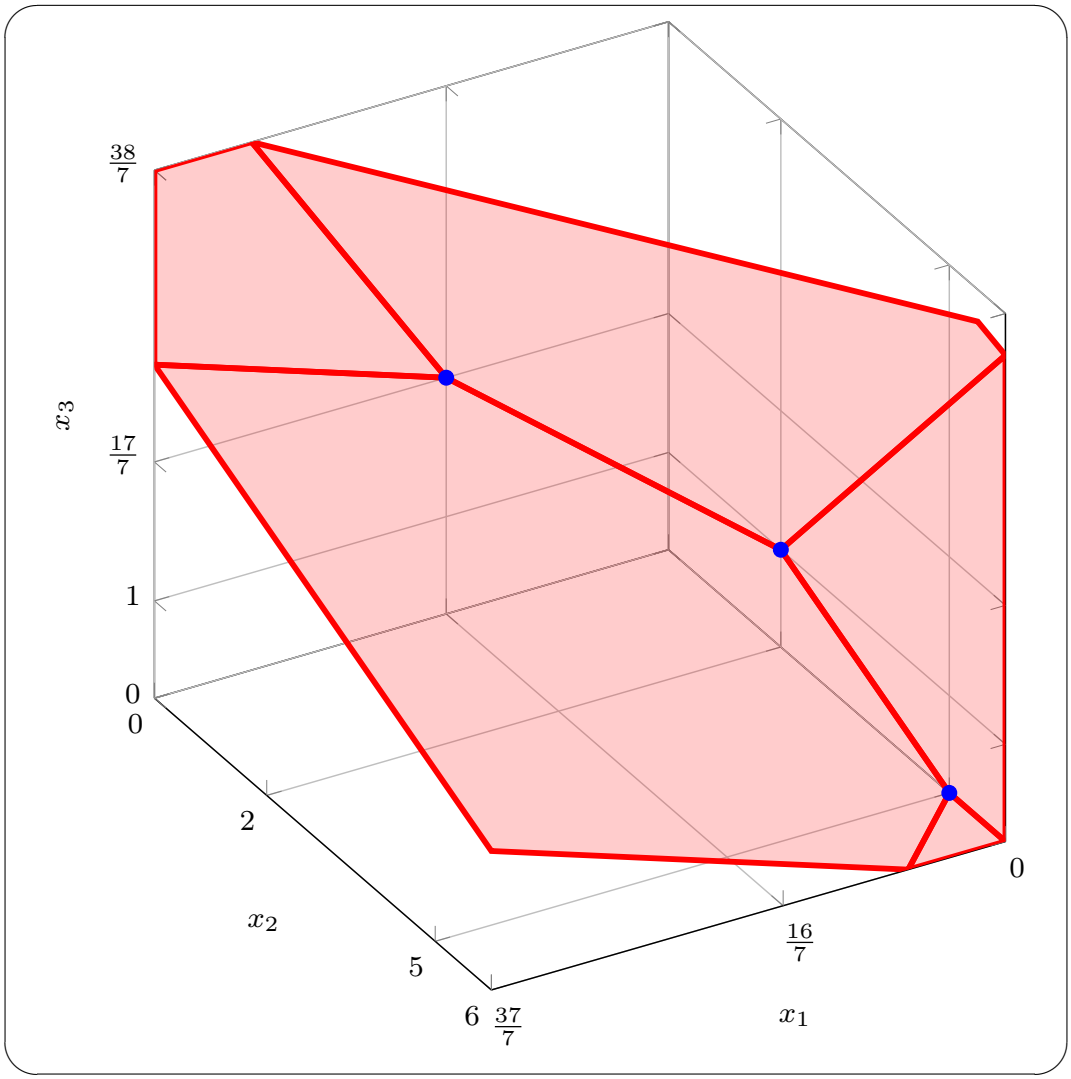


Capitolo 6

Problemi LP con tre variabili – Tipo B

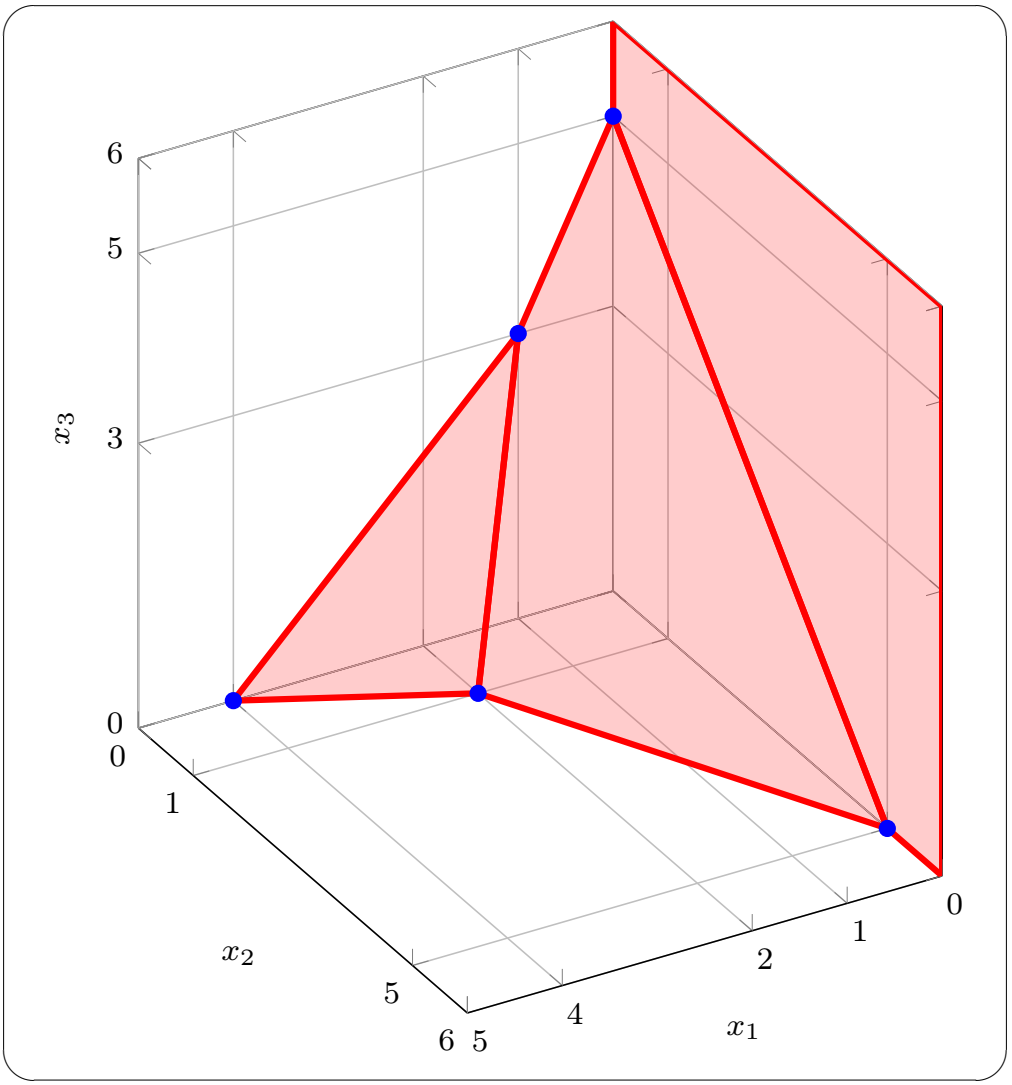
6.1 Problema LP *P*51

$$
\begin{array}{lrcrcrcrcr}
\max & - & 2\,x_1 & - & x_2 & & & & & \\
\text{soggetto a} & & x_1 & - & x_2 & - & 3 & x_3 & \leq & -5 \\
& - & 3 & x_1 & - & 2 & x_2 & + & 2 & x_3 & \leq & -2 \\
& & x_1, & & & x_2, & & & x_3 & \geq & 0
\end{array}
$$



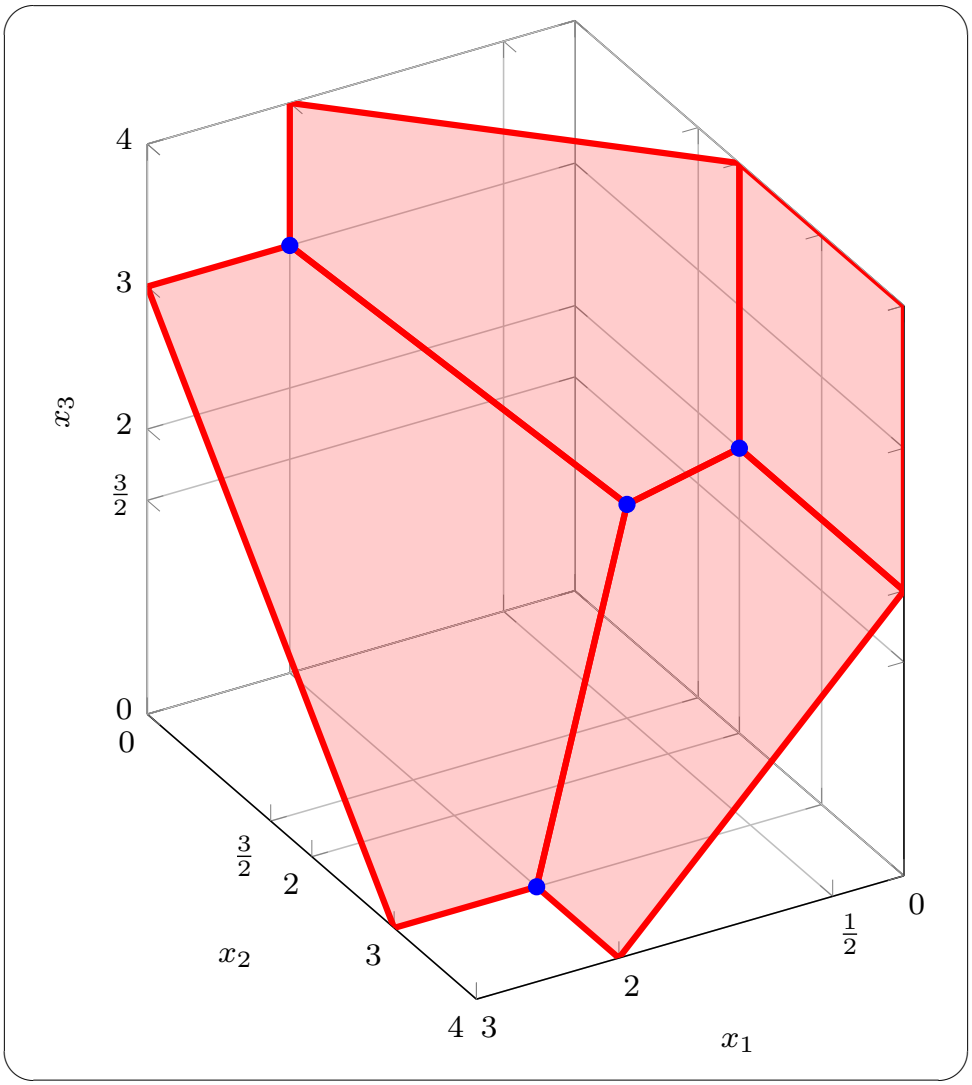
6.2 Problema LP *P52*

$$
\begin{array}{rrrrrrrrrr}
\max & - & x_1 & - & 2\,x_2 & - & x_3 & & \\
\text{soggetto a} & - & 2\,x_1 & - & x_2 & - & x_3 & \leq & -5 \\
& - & x_1 & - & 2\,x_2 & - & x_3 & \leq & -4 \\
& - & x_1 & - & x_2 & - & 2\,x_3 & \leq & -3 \\
& & x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$



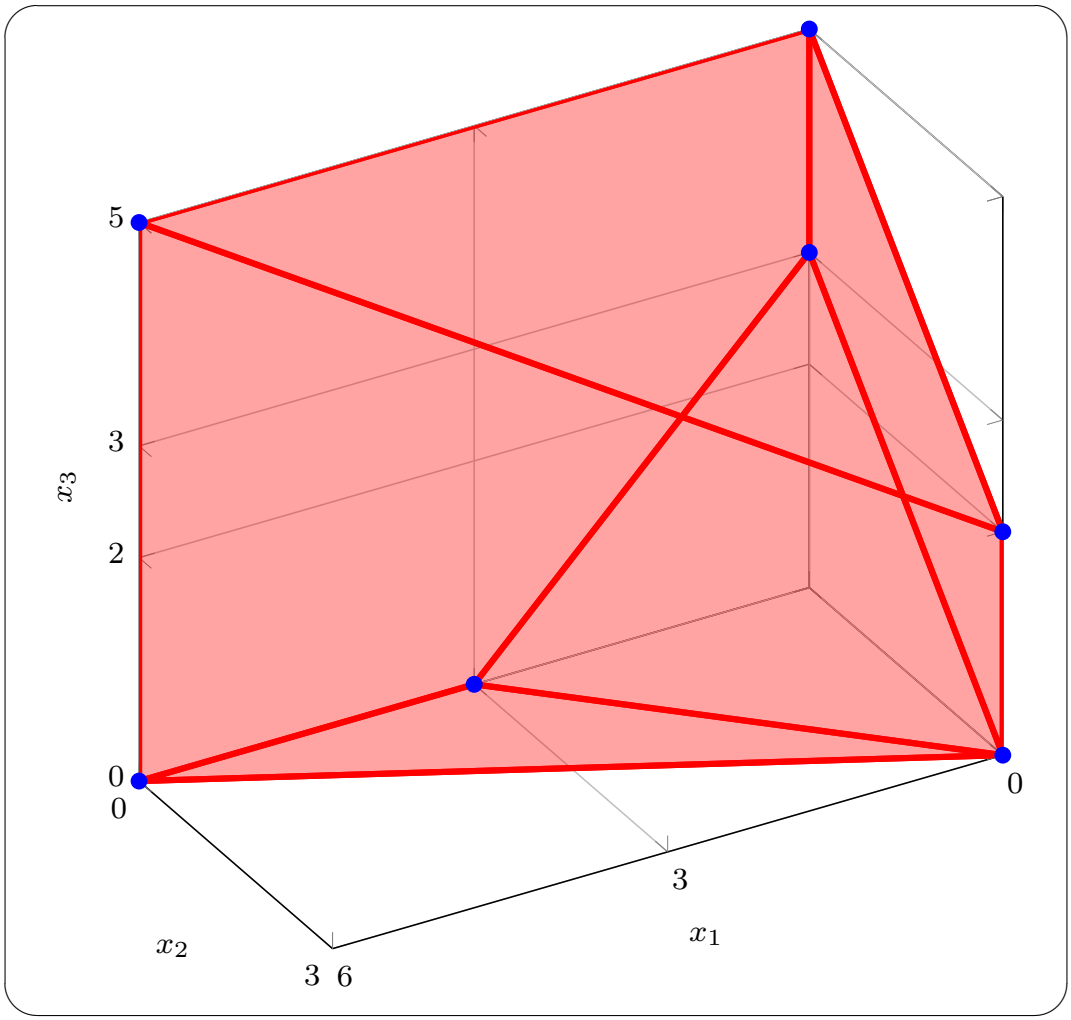
6.3 Problema LP *P*53

$$
\begin{array}{rrrrrrrrrr}
\max & - & x_1 & - & 2\,x_2 & - & 4\,x_3 & & \\
\text{soggetto a} & - & x_1 & - & x_2 & & & \leq & -2 \\
 & & & - & x_2 & - & x_3 & \leq & -3 \\
 & - & x_1 & & & - & x_3 & \leq & -2 \\
 & & x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$



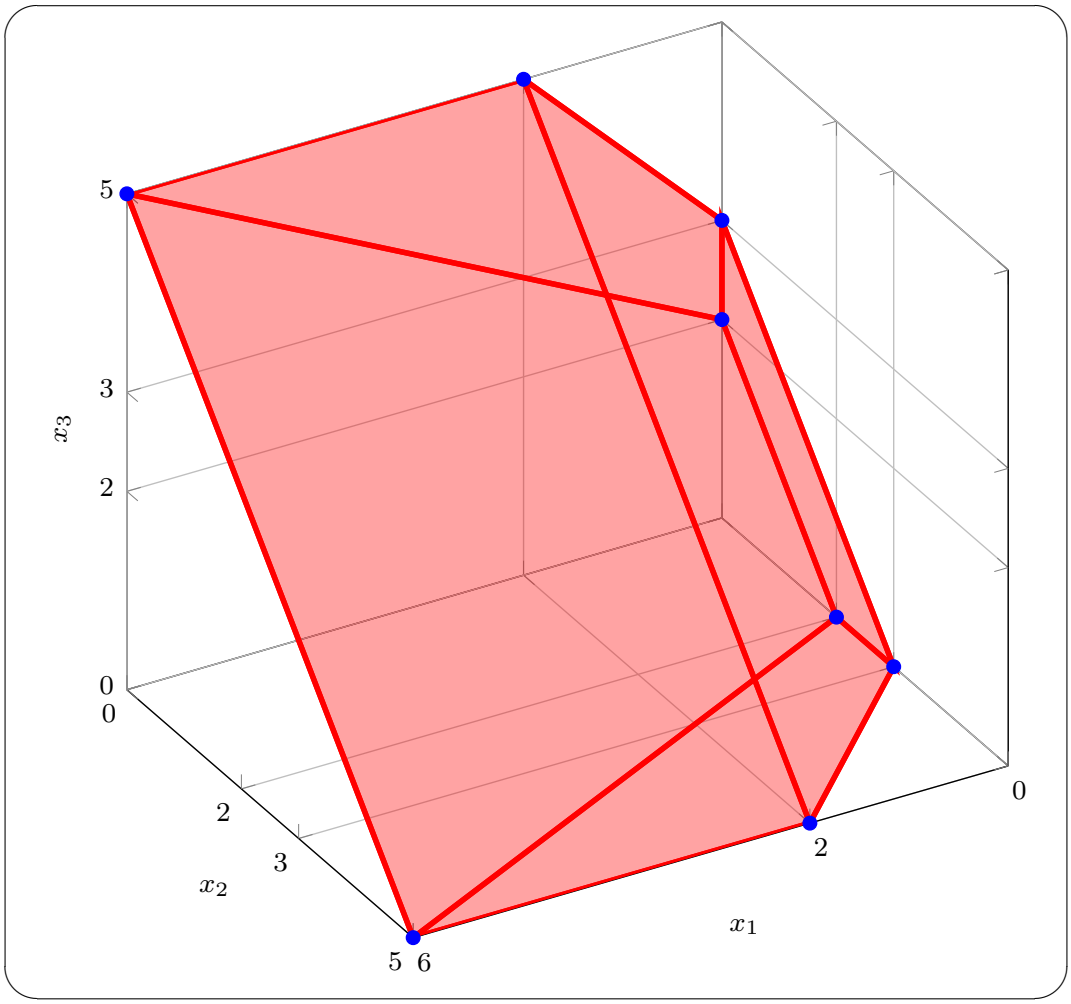
6.4 Problema LP *P54*

$$
\begin{array}{rrrrrrrrr}
\max & - & x_1 & - & 2\,x_2 & - & 2\,x_3 & & \\
\text{soggetto a} & & x_1 & + & 2\,x_2 & & & \leq & 6 \\
& - & x_1 & - & x_2 & - & x_3 & \leq & -3 \\
& & & & x_2 & + & x_3 & \leq & 5 \\
& & x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$



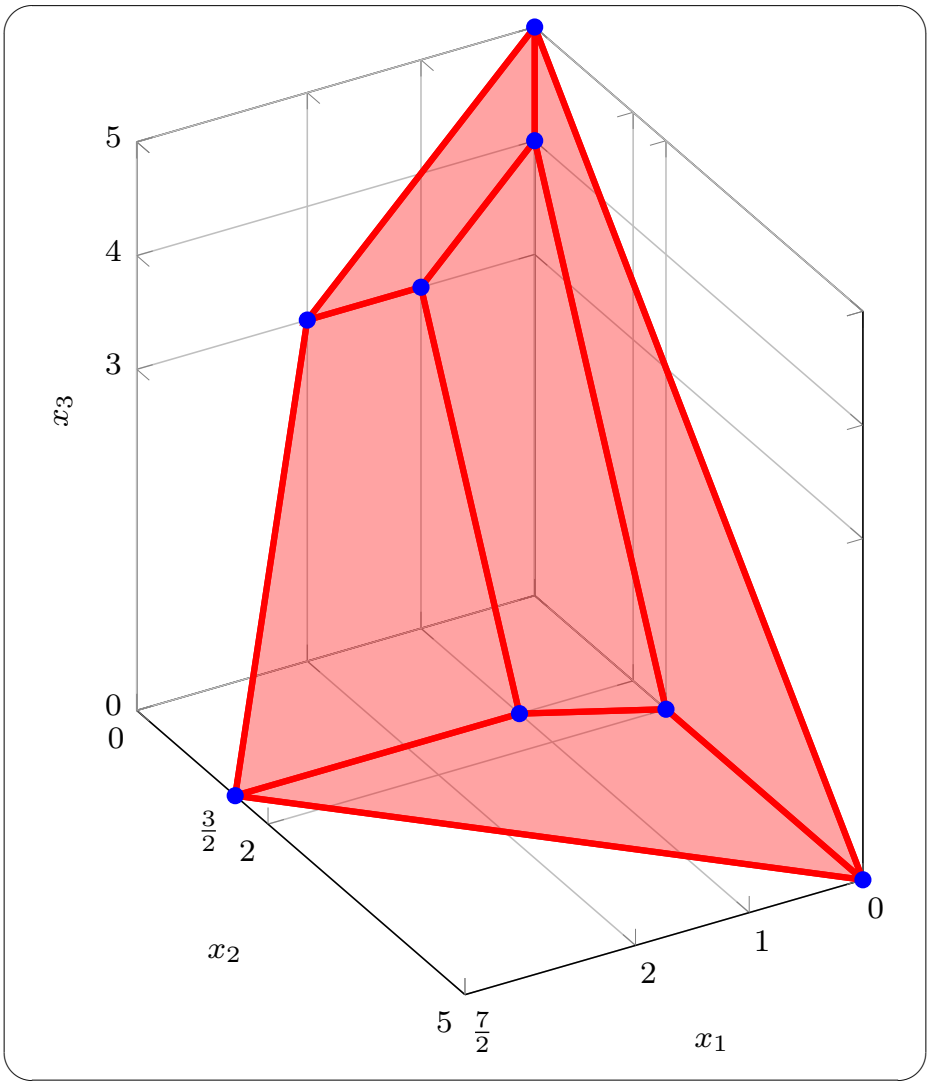
6.5 Problema LP *P*55

$$
\begin{array}{rrrrrrrrr}
\max & - & 2\,x_1 & - & 2\,x_2 & - & x_3 & & \\
\text{soggetto a} & & & & x_2 & + & x_3 & \leq & 5 \\
& - & x_1 & + & x_2 & + & x_3 & \leq & 3 \\
& & x_1 & - & 2\,x_2 & - & 2\,x_3 & \leq & -4 \\
& & x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$



6.6 Problema LP *P56*

$$
\begin{array}{lrcrcrcr}
\max & -2x_1 & - & 3x_2 & - & 3x_3 & & \\
\text{soggetto a} & -x_1 & - & 2x_2 & - & x_3 & \leq & -4 \\
 & & - & 2x_2 & - & x_3 & \leq & -3 \\
 & x_1 & + & x_2 & + & x_3 & \leq & 5 \\
 & x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$

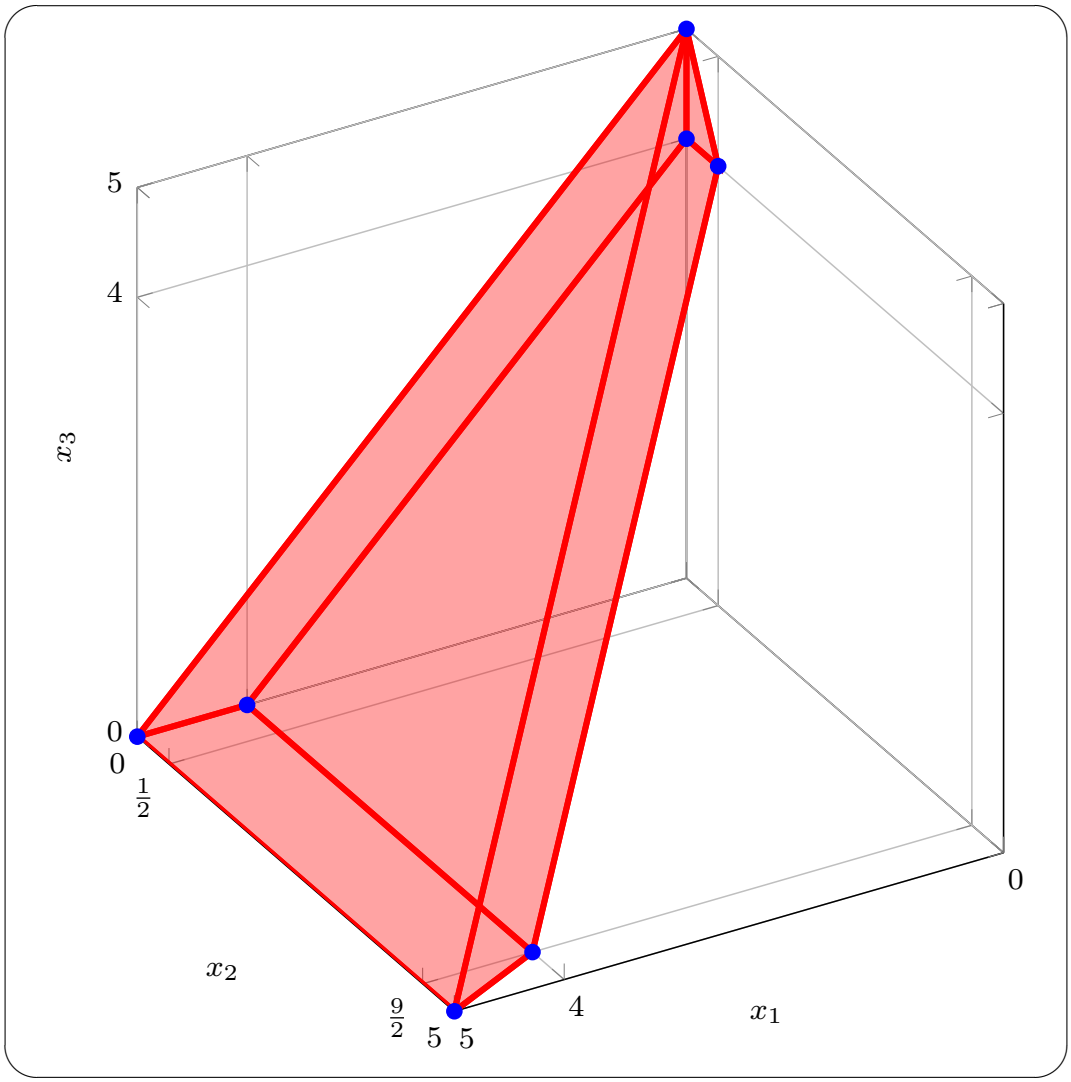


Capitolo 7

Problemi LP con tre variabili – Tipo C

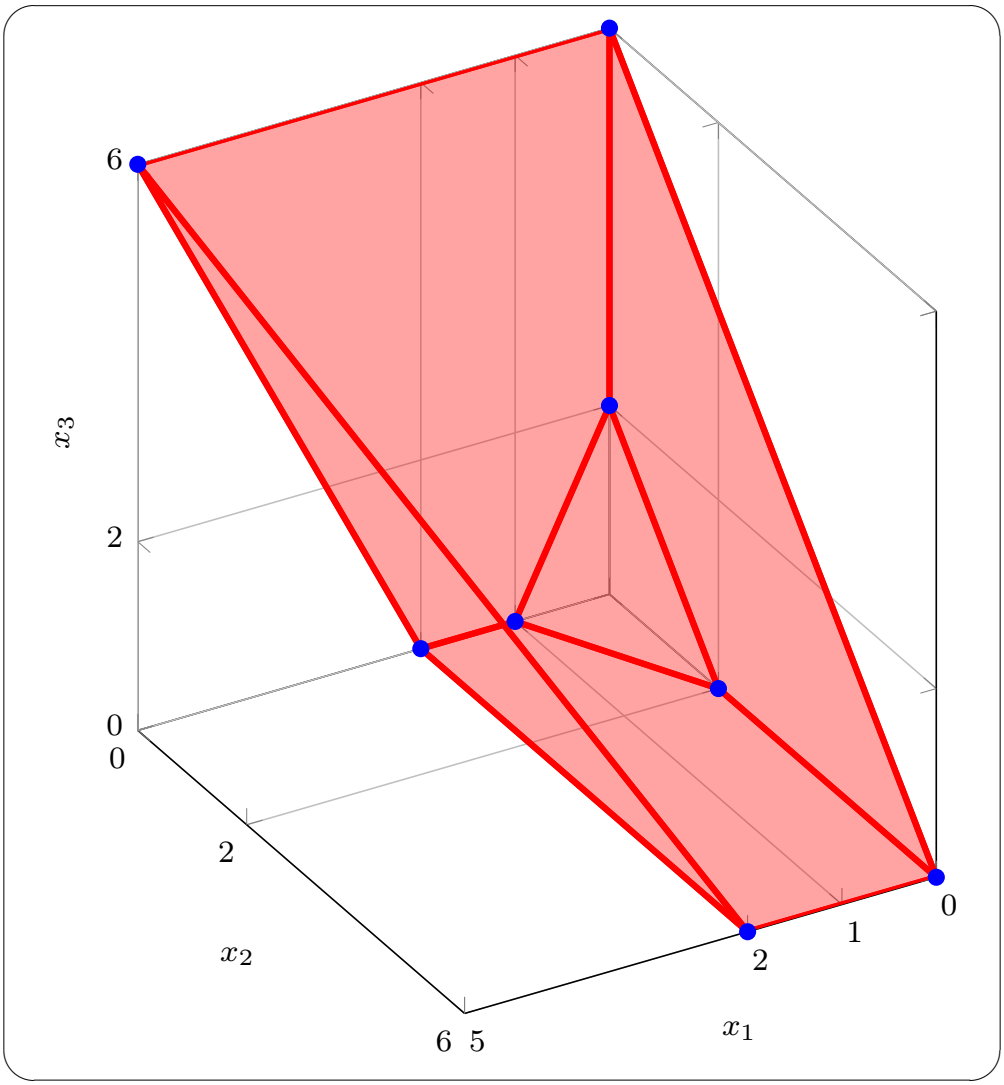
7.1 Problema LP *P61*

$$
\begin{array}{rrcrcrcr}
\max & 3\,x_1 & + & x_2 & + & 2\,x_3 & & \\
\text{soggetto a} & x_1 & & & + & x_3 & \leq & 5 \\
- & x_1 & & & - & x_3 & \leq & -4 \\
- & x_1 & + & 2\,x_2 & + & x_3 & \leq & 5 \\
& x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$



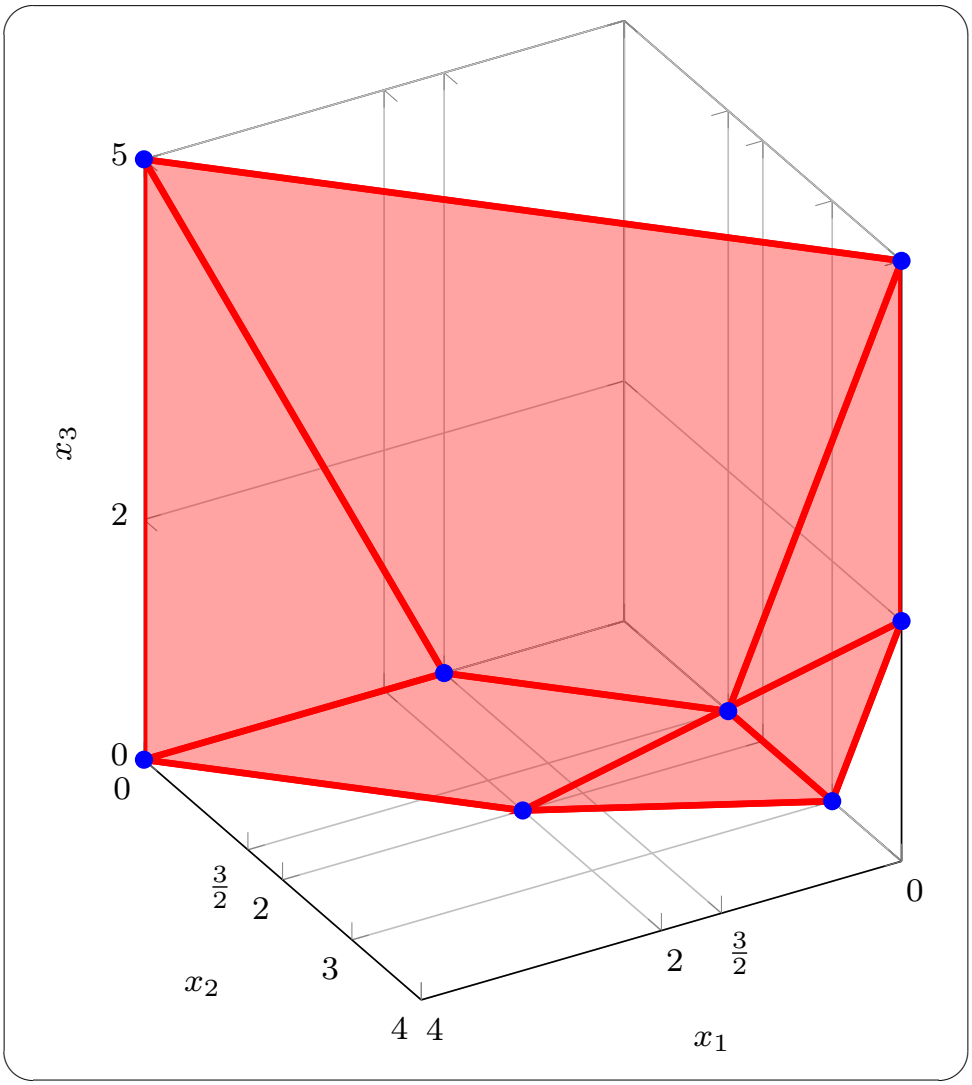
7.2 Problema LP *P62*

$$
\begin{array}{rrcrcrcr}
\max & 2\,x_1 & + & 3\,x_2 & + & x_3 & & \\
\text{soggetto a} & & & x_2 & + & x_3 & \leq & 6 \\
- & 2\,x_1 & - & x_2 & - & x_3 & \leq & -2 \\
& 2\,x_1 & & & - & x_3 & \leq & 4 \\
& x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$



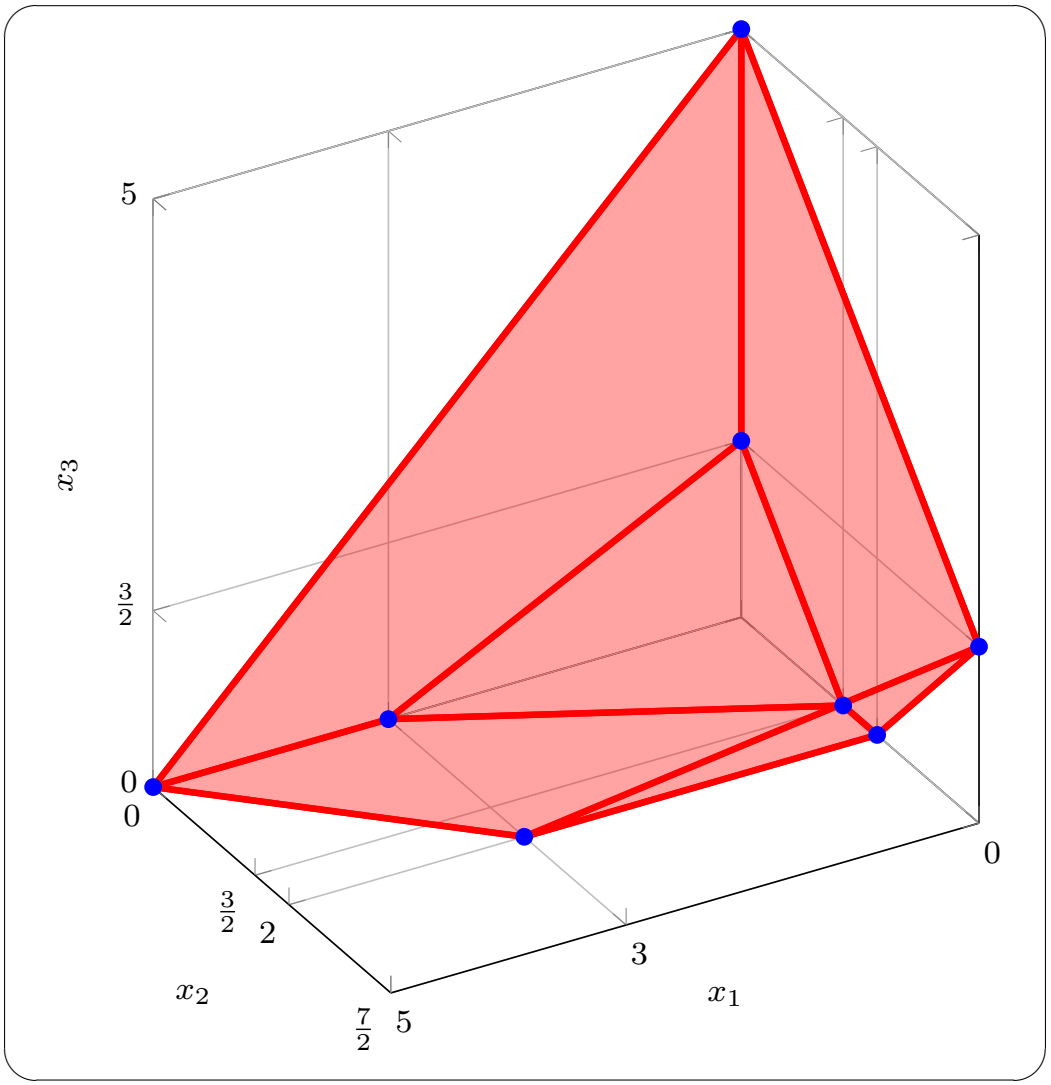
7.3 Problema LP *P63*

$$
\begin{array}{rrcrcrcr}
\max & x_1 & + & 2\,x_2 & + & 2\,x_3 & & \\
\text{soggetto a} & x_1 & + & x_2 & & & \leq & 4 \\
 & x_1 & + & 2\,x_2 & - & x_3 & \leq & 6 \\
- & 2\,x_1 & - & 2\,x_2 & + & x_3 & \leq & -3 \\
 & x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$



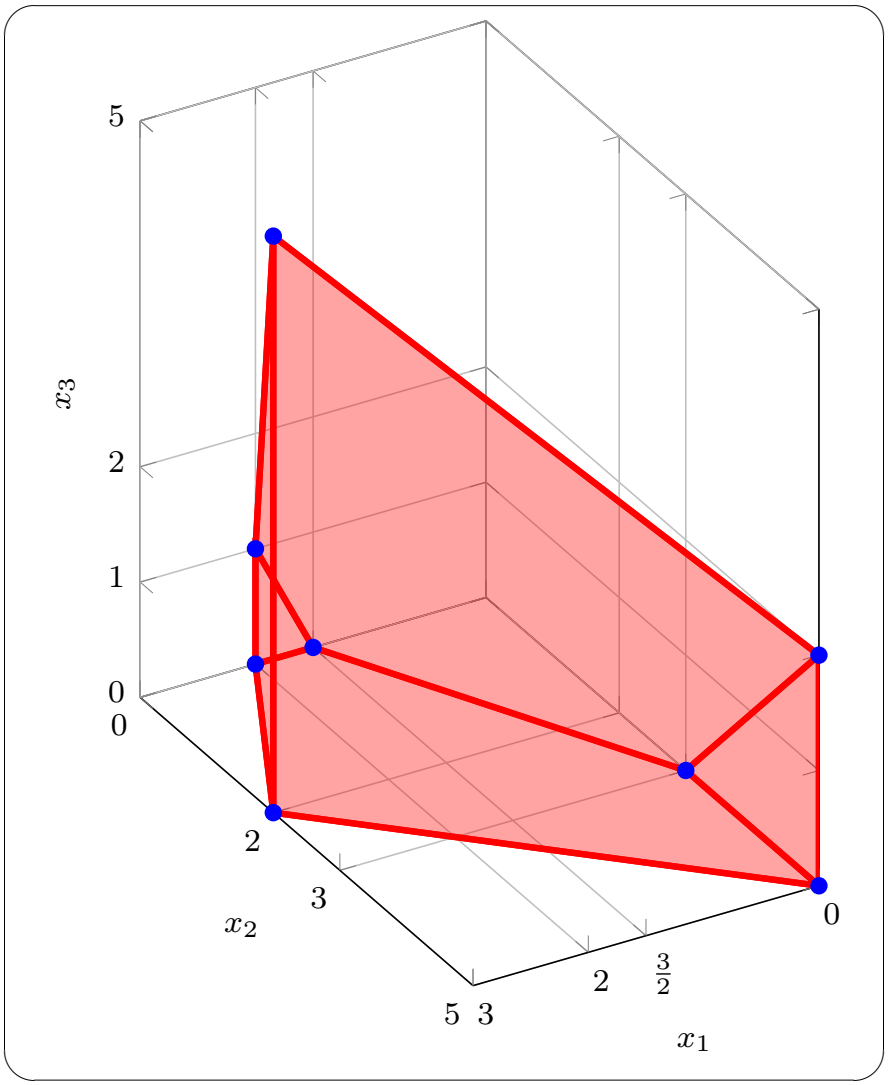
7.4 Problema LP *P64*

$$
\begin{array}{lrcrcrcr}
\max & 3\,x_1 & + & 2\,x_2 & + & x_3 & & \\
\text{soggetto a} \quad - & x_1 & - & 2\,x_2 & - & 2\,x_3 & \leq & -3 \\
& x_1 & + & x_2 & + & x_3 & \leq & 5 \\
& & & x_2 & - & x_3 & \leq & 2 \\
& x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$



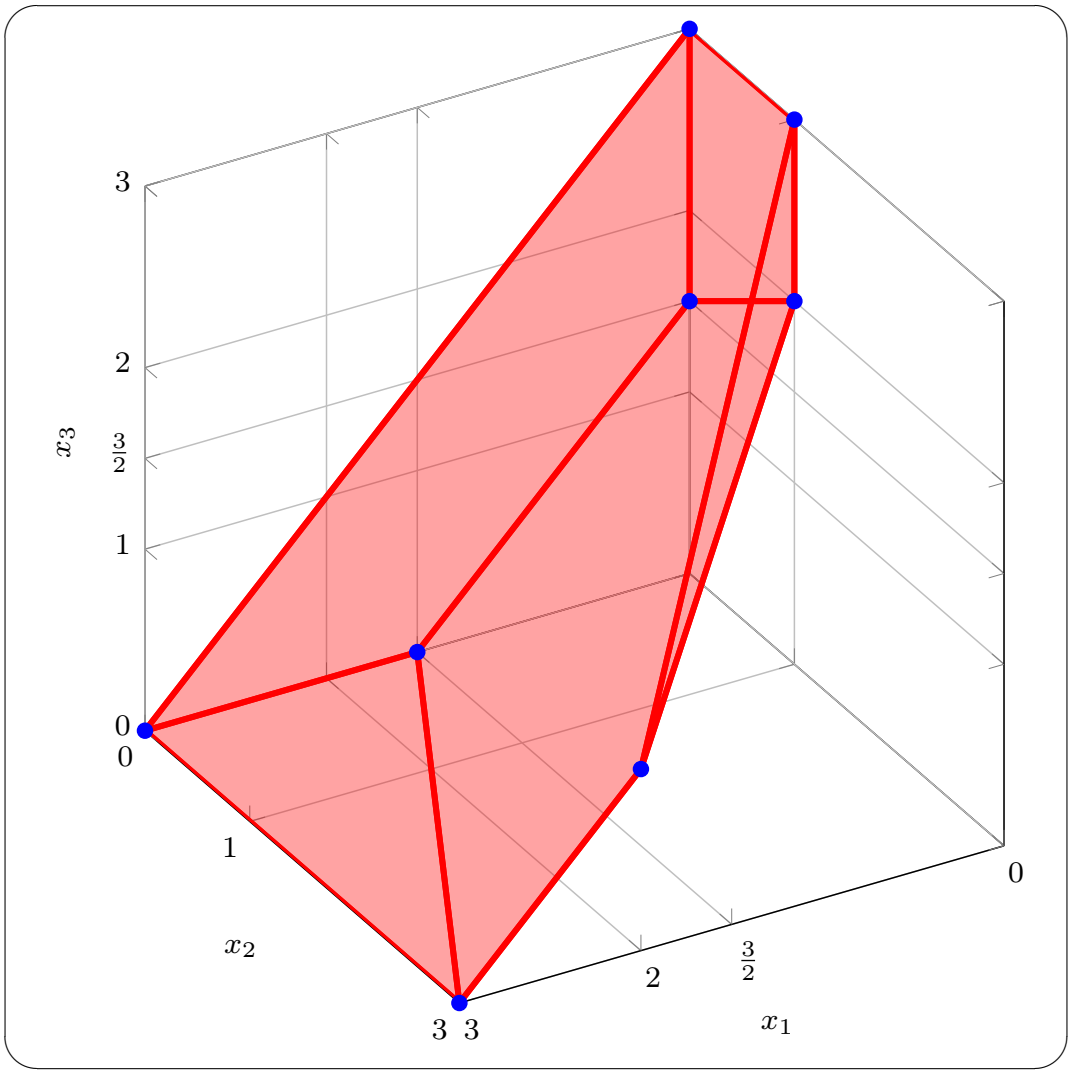
7.5 Problema LP *P65*

$$
\begin{array}{rrcrcrcr}
\max & 2\,x_1 & + & 3\,x_2 & + & 3\,x_3 & & \\
\text{soggetto a} & x_1 & + & x_2 & & & \leq & 5 \\
& 2\,x_1 & - & x_2 & & & \leq & 4 \\
- & 2\,x_1 & - & x_2 & + & x_3 & \leq & -3 \\
& x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$



7.6 Problema LP *P66*

$$
\begin{array}{lrcrcrcr}
\max & 2\,x_1 & + & x_2 & + & x_3 & & \\
\text{soggetto a} & -\,2\,x_1 & + & x_2 & - & 2\,x_3 & \leq & -3 \\
& x_1 & & & + & x_3 & \leq & 3 \\
& -\,2\,x_1 & + & 2\,x_2 & & & \leq & 2 \\
& x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$

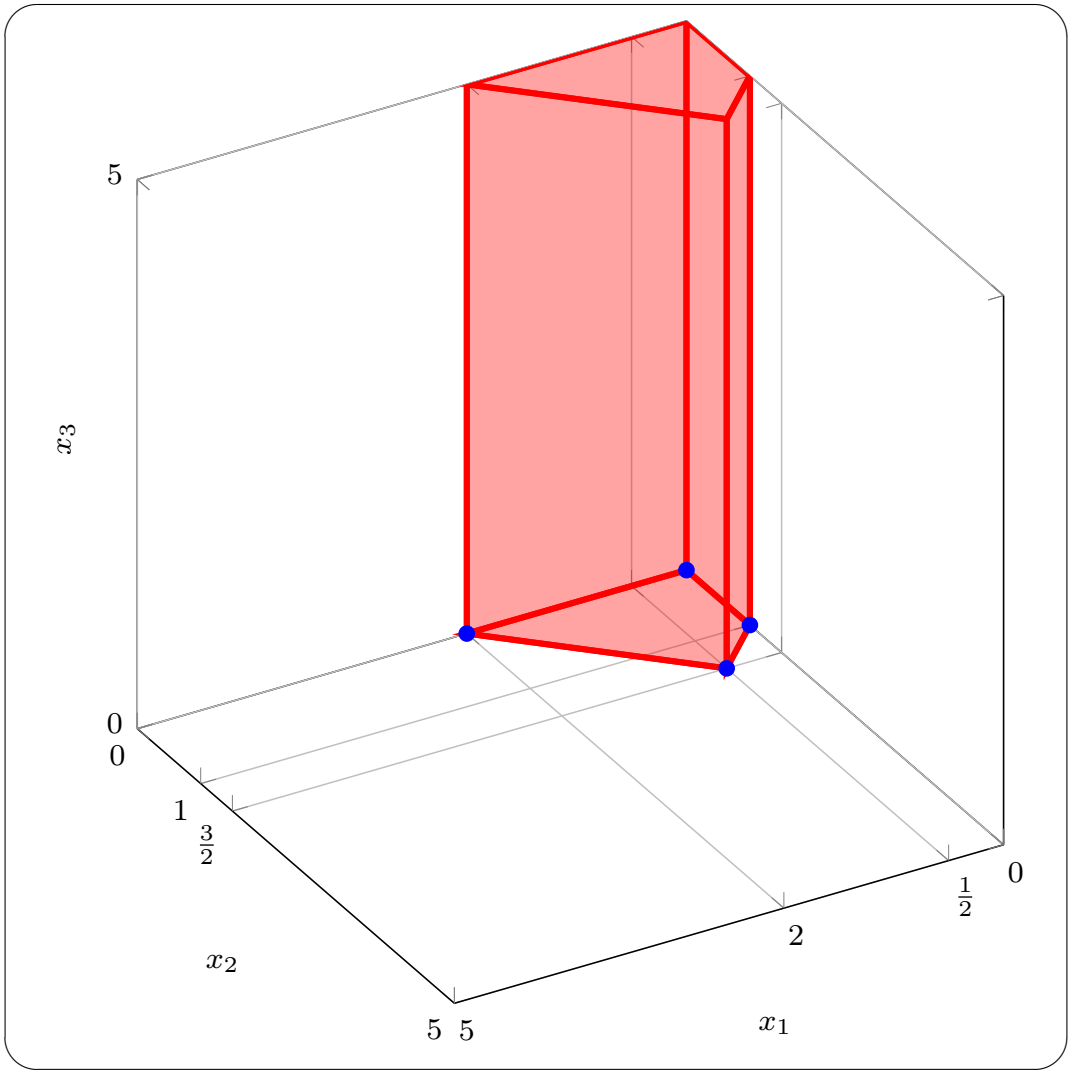


Capitolo 8

Problemi LP con tre variabili – Tipo D

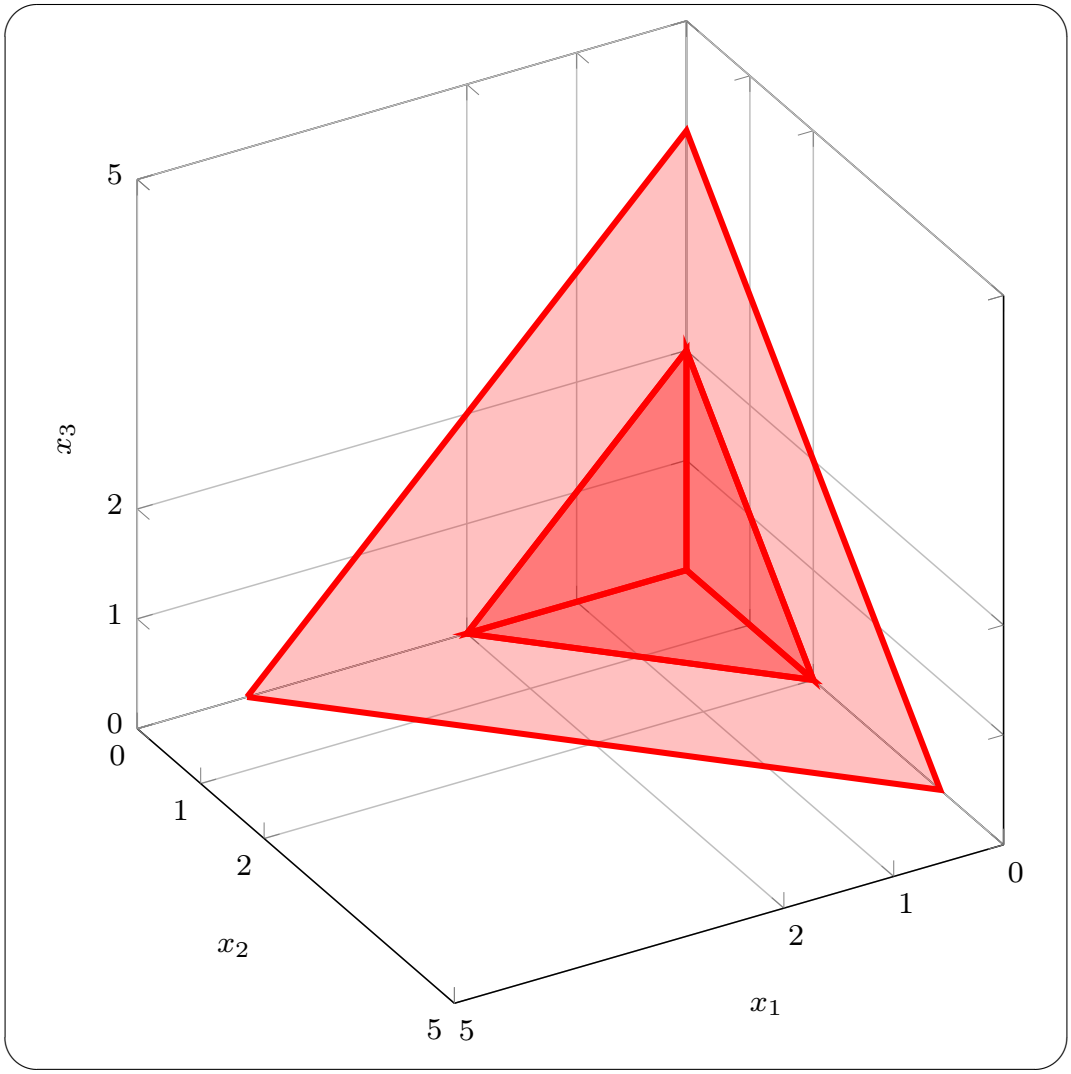
8.1 Problema LP *P81*

$$
\begin{array}{lrcrcrcl}
\max & & x_1 & + & x_2 & + & x_3 & \\
\text{soggetto a} & & x_1 & + & x_2 & & & \leq 2 \\
& - & x_1 & + & x_2 & & & \leq 1 \\
& & x_1, & & x_2, & & x_3 & \geq 0
\end{array}
$$



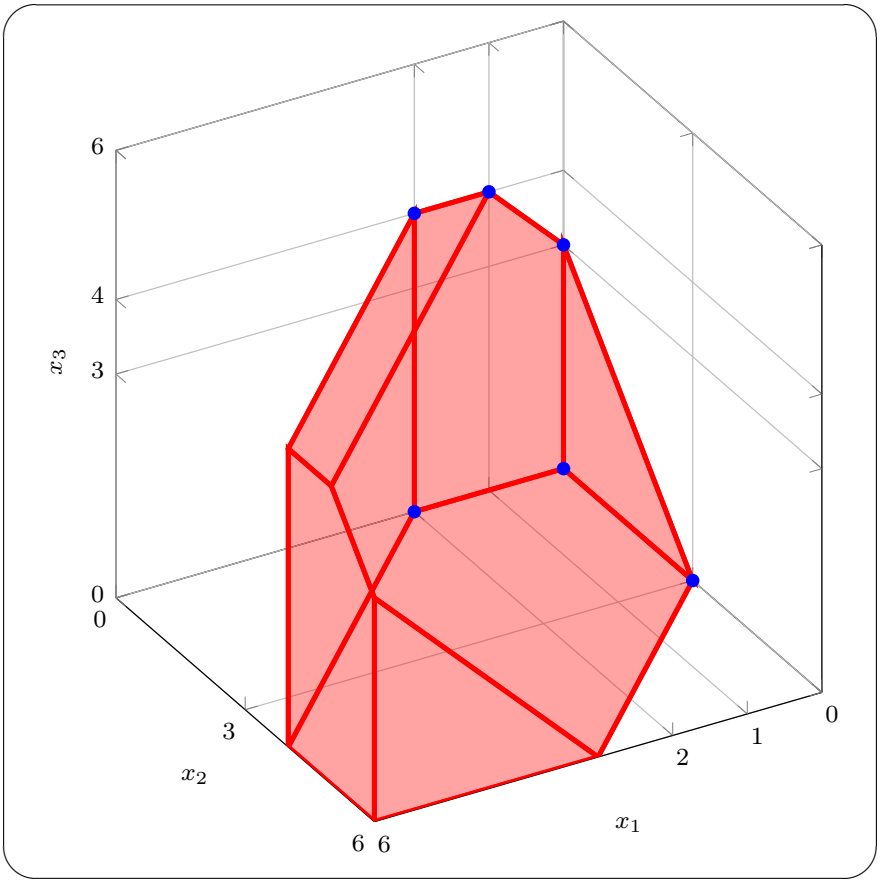
8.2 Problema LP *P82*

$$
\begin{array}{rrcrcrcrl}
\max & & x_1 & + & x_2 & + & x_3 & & \\
\text{soggetto a} & & x_1 & + & x_2 & + & x_3 & \leq & 2 \\
& - & x_1 & - & x_2 & - & x_3 & \leq & -4 \\
& & x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$



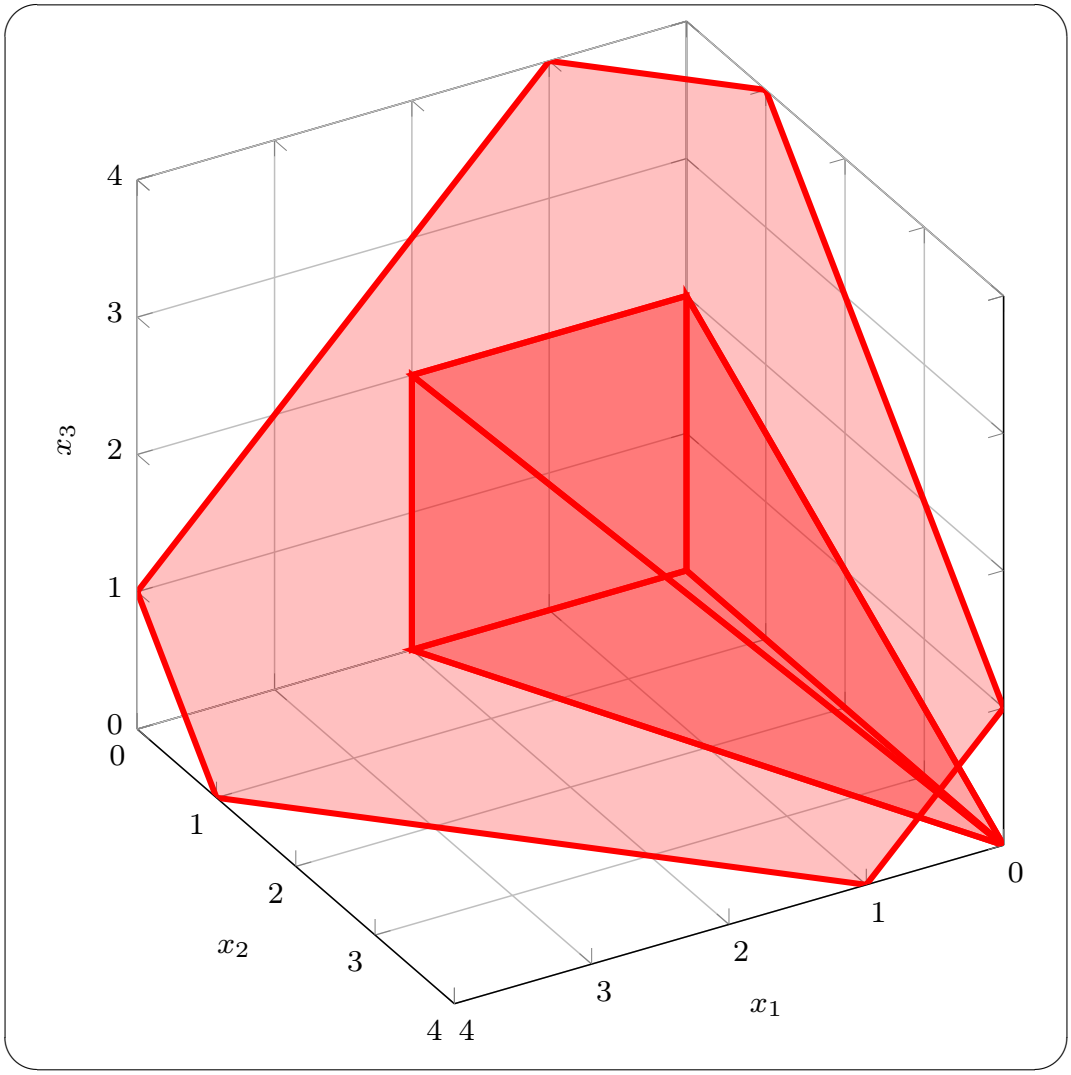
8.3 Problema LP *P83*

$$
\begin{array}{lrcrcrcl}
\max & x_1 & + & 2\,x_2 & + & x_3 & & \\
\text{soggetto a} & x_1 & - & x_2 & & & \leq & 2 \\
- & x_1 & + & x_2 & + & x_3 & \leq & 3 \\
 & & & & & x_3 & \leq & 4 \\
 & x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$



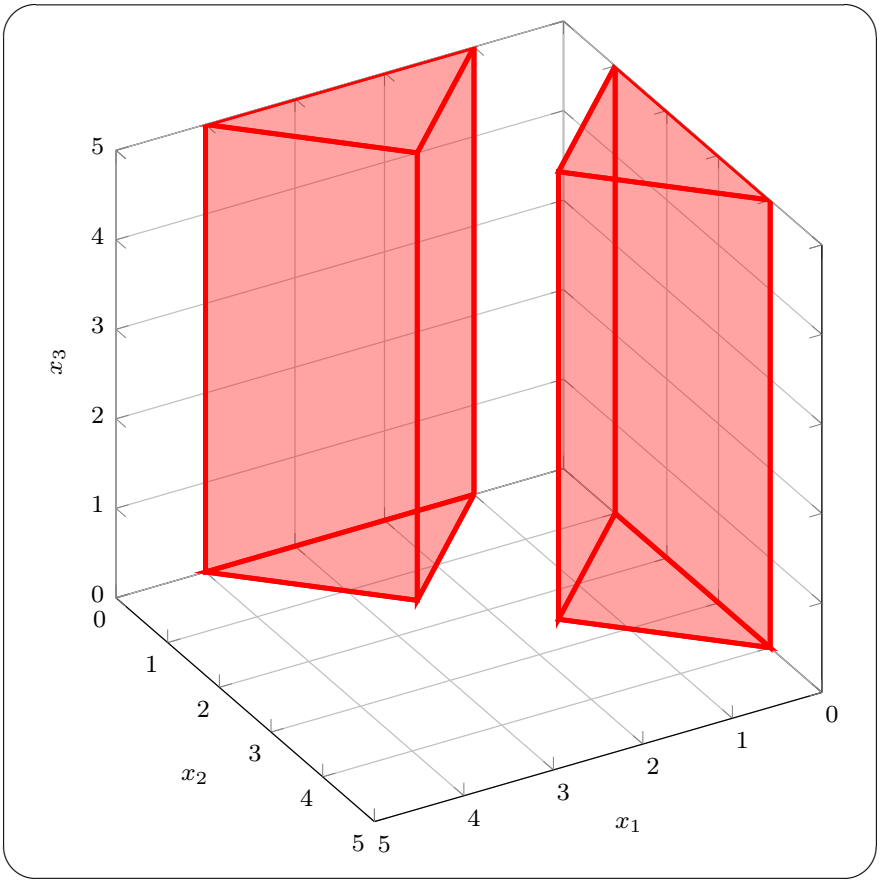
8.4 Problema LP *P84*

$$
\begin{array}{rrcrcrcr}
\max & x_1 & + & x_2 & + & x_3 & & \\
\text{soggetto a} & 2\,x_1 & + & x_2 & & & \leq & 4 \\
 & & & x_2 & + & 2\,x_3 & \leq & 4 \\
- & x_1 & - & x_2 & - & x_3 & \leq & -5 \\
 & x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$



8.5 Problema LP *P85*

$$
\begin{array}{lrcrcrcr}
\max & & x_1 & + & x_2 & + & x_3 & \\
\text{soggetto a} & & x_1 & - & x_2 & & \leq & -1 \\
& - & x_1 & + & x_2 & & \leq & -1 \\
& & x_1 & + & x_2 & & \leq & 4 \\
& & x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$



8.6 Problema LP *P86*

$$
\begin{array}{lrcrcrcr}
\max & 2\,x_1 & + & 2\,x_2 & - & x_3 & & \\
\text{soggetto a} & x_1 & + & x_2 & & & \leq & 4 \\
 & x_1 & & & - & x_3 & \leq & 1 \\
- & x_1 & - & x_2 & & & \leq & -1 \\
 & x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$

